

CHEMISTRY 201



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Chemistry 201-- Professor Samar Ayesb

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Licensing

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CHAPTER OVERVIEW

1: Units of Measurement

Introduction to Module

Measurement is a fundamental part of everyday life and of all scientific work. Whether we are calculating distance, measuring ingredients for a recipe, recording temperature, or analyzing data in a laboratory, we rely on standardized units to describe quantities clearly and consistently. Units of measurement allow us to communicate numerical information accurately and ensure that results can be understood and replicated anywhere in the world.

A **measurement** consists of two parts: a number and a unit. The number tells us how much of something there is, and the unit tells us what standard is being used. For example, “5 meters” combines the number 5 with the unit meter to describe length. Without units, numbers have little meaning and can easily lead to confusion or error.

Most countries use the **International System of Units (SI)**, a standardized system based on seven base units. These include the meter (length), kilogram (mass), second (time), ampere (electric current), kelvin (temperature), mole (amount of substance), and candela (luminous intensity). From these base units, many other units—called derived units—are formed, such as meters per second for speed or newtons for force.

In addition to SI units, other measurement systems are still used in certain contexts, such as the U.S. customary system (inches, pounds, gallons). Understanding how to convert between systems is an important skill in science, engineering, healthcare, and daily life.

Accurate measurement also requires attention to precision and significant figures. Instruments have limitations, and measurements always involve some degree of uncertainty. Recognizing this uncertainty is essential for interpreting data responsibly and making informed conclusions.

In this module, you will learn how units are defined, how to convert between them, how to apply dimensional analysis, and how measurement supports scientific reasoning. Mastery of units is not just a mathematical skill—it is a foundational tool for understanding the physical world.

Learning Outcomes

By the end of this module, students are expected to:

- Identify common SI base units and derived units used in chemistry.
- Interpret and apply metric prefixes when expressing measurements.
- Convert quantities within the metric system using powers of ten.
- Apply dimensional analysis to convert between units.
- Perform multi-step unit conversions involving metric and English units.
- Describe the Celsius, Kelvin, and Fahrenheit temperature scales.
- Convert temperatures between scales.
- Define density as the ratio of mass to volume.
- Calculate density, mass, or volume when two variables are known.
- Identify the correct number of significant figures in measured values.
- Apply significant figure rules in calculations.
- Report numerical results with appropriate precision and rounding.
- Distinguish between measurement accuracy and precision.

[1.1: Units of Measurement--Readings](#)

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1.1: Units of Measurement--Readings

Metric Units

Scientific measurements all over the world, except in the United States, are made using the **metric system**. Scientists prefer the metric system because it is internationally standardized. Scientists adopted the metric system for their standard of measurement because it is decimal-based. Units differ by powers of ten. There are 10 millimeters in a centimeter, 10 centimeter in a decimeter, and 10 decimeter in a meter. Compare that to the English system used in the US, where 12 inches per foot, 3 feet per yard, 1760 yards per mile, and so on, are used.

Modern scientists use the **SI units**, which are a subset of units in the metric system. SI is an abbreviation for the French name for the International System of Units. The SI system is based on seven **fundamental units**, or **base units**, each identified with a physical quantity (Table 1.1). Other quantities are made up of combinations of base units; these are called **derived units**. In this section, we will describe three of the base units-units of mass, length, and temperature. Two of the derived units, units for volume and density, will be described here as well.

Table 1.1 SI Base Units		
Quantity	Unit	Symbol
Length	Meter	m
Mass	Kilogram	Kg
Time	Second	s
Temperature	Kelvin	K
Amount of Substance	Mole	mol
Electric Current	Ampere	A
Luminous Intensity	Candela	cd

For quantities much smaller or larger than the base unit, we use decimal prefixes and exponential (scientific) notation (Table 1.2). Because the prefixes are based on powers of 10, SI units are easier to use in calculations than English units.

Table 1.2 Common Decimal Prefixes Used with SI Units

Prefix*	Prefix Symbol	Word	Conventional Notation	Exponential Notation
tera	T	trillion	1,000,000,000,000	1×10^{12}
giga	G	billion	1,000,000,000	1×10^9
mega	M	million	1,000,000	1×10^6
kilo	k	thousand	1,000	1×10^3
--	--	one	1	1×10^1
deci	d	tenth	0.1	1×10^{-1}
centi	c	hundredth	0.01	1×10^{-2}
milli	m	thousandth	0.001	1×10^{-3}
micro	μ	millionth	0.000001	1×10^{-6}
nano	n	billionth	0.000000001	1×10^{-9}
pico	p	trillionth	0.000000000001	1×10^{-12}
femto	f	quadrillionth	0.000000000000001	1×10^{-15}

*The prefixes most frequently used by chemists appear in bold type.

Important SI Units in Chemistry

In this section we will discuss units for length, mass, temperature, volume and density. Table 1.3 shows some SI quantities for length, volume, and mass, along with their English-system equivalents.

Length The SI unit of length is the **meter (m)**. The definition is exact and invariant: 1 meter is the distance light travels in a vacuum in $1/299,792,458$ of a second. A meter is a little longer than a yard ($1 \text{ m} = 1.094 \text{ yd}$). Biological cells are often measured in micrometers ($1 \mu\text{m} = 10^{-6} \text{ m}$). On the atomic scale, nanometers (10^{-9} m) and picometers (10^{-12} m) are used. Many proteins have diameters of about 2 nm; atomic diameters are about 200 pm (0.2 nm). An older unit still in use is the angstrom ($1 \text{ \AA} = 10^{-10} \text{ m} = 0.1 \text{ nm} = 100 \text{ pm}$).

Table 1.3 Common SI-English Equivalent Quantities

Quantity	SI to English Equivalent	English to SI Equivalent
Length	1 km = 0.6214 mile (mi) 1 m = 1.094 yards (yd) 1 m = 39.37 inches (in) 1 cm = 0.3937 inch	1 mile = 1.609 km 1 yd = 0.9144 m 1 foot (ft) = 0.3048 m 1 inch = 2.54 cm (exactly)
Volume	1 cubic meter (m^3) = 35.31 ft^3 1 cubic decimeter (dm^3) = 0.2642 gallon (gal) 1 dm^3 = 1.057 quart (qt) 1 cubic centimeter (cm^3) = 0.03381 fluid ounce	1 ft^3 = 0.02832 m^3 1 gal = 3.785 dm^3 1 qt = 0.9464 dm^3 1 qt = 946.4 cm^3 1 fluid ounce = 29.57 cm^3
Mass	1 kg = 2.205 pounds (lb) 1 g = 0.03527 ounce (oz)	1 lb = 0.4536 kg 1 oz = 28.35 g

Mass The quantity of matter an object contains is its **mass**. The SI unit of mass is the **kilogram (kg)**. The terms *mass* and *weight* have distinct meanings:

* **Mass is constant** because an object's quantity of matter cannot change.

* **Weight is variable** because it depends on the local gravitational field.

Because the strength of gravitational field varies with altitude, you (and other objects) weigh slightly less on a high mountain than at sea level.

Temperature is a measure of how hot or cold one object is relative to another. In the laboratory, we measure temperature with a thermometer. We'll consider three temperature scales: the Celsius (°C); the Kelvin (K), and the Fahrenheit (°F) scales. The SI base unit of temperature is the **kelvin (K)**, with no degree sign, °. In the United States, the Fahrenheit scale is the one generally used.

The three scales differ in the size of the unit and/or the temperature of the zero point. Figure 1.1 shows the freezing and boiling points of water in the three scales. The **Celsius scale** set water's freezing point at 0°C and its boiling point (at normal atmospheric pressure) at 100°C. The **Kelvin (absolute) scale** uses the same size degree as the Celsius scale, but it has a different zero point; that is 0 K, or *absolute zero*, equals -273.15°C. Thus, in the Kelvin scale, all temperatures are positive.

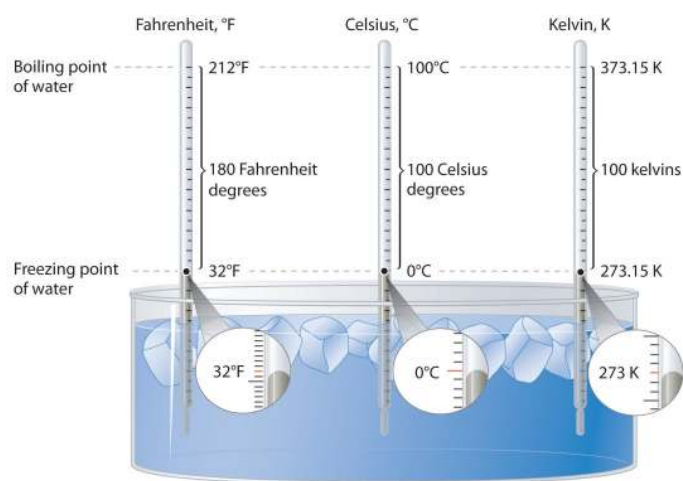


Figure 1.1 Freezing and boiling points of water in the Fahrenheit, Celsius, and Kelvin (absolute) scales. Because the difference between the freezing point of water and the boiling point of water is 100° on both the Celsius and Kelvin scales, the size of a degree Celsius (°C) and a kelvin (K) are precisely the same. In contrast, both a degree Celsius and a kelvin are 9/5 the size of a degree Fahrenheit (°F). https://saylordotorg.github.io/text_...-skills-4.html

We convert between Celsius and Kelvin scales by using the following equations:

$$T \text{ (in K)} = T \text{ (in } ^\circ\text{C)} + 273.15$$

And therefore,

$$T \text{ (in } ^\circ\text{C)} = T \text{ (in K)} - 273.15$$

The Fahrenheit scale differs from the other scales in its zero point **and** in the size of its degree. Water freezes at 32°F and boils at 212°F. Therefore, 180 Fahrenheit degrees (212°F - 32°F) represents the same temperature change as 100 Celsius degrees or (100 kelvins).

To convert a temperature from °C to °F, we use the following equation:

$$T \text{ (in } ^\circ\text{F)} = \frac{9}{5} T \text{ (in } ^\circ\text{C)} + 32$$

To convert a temperature from °F to °C, we use the following equation:

$$T \text{ (in } ^\circ\text{C)} = [T \text{ (in } ^\circ\text{F)} - 32] \times \frac{5}{9}$$

Volume The amount of space that any sample of matter occupies is defined as the volume. The SI unit of volume is the **cubic meter (m³)**. In chemistry, we often use the non-SI units **liter (L)** and **milliliter (mL)**. Medical practitioners measure body fluids in

cubic decimeters (dm³), which are equivalent to liters:

$$1 \text{ L} = 1 \text{ dm}^3 = 10^{-3} \text{ m}^3 \text{ and,}$$

$$1 \text{ mL} = 1 \text{ cm}^3 = 10^{-3} \text{ dm}^3 = 10^{-3} \text{ L} = 10^{-6} \text{ m}^3$$

A liter is slightly larger than a quart (1 L = 1.057 qt; 1 qt = 946.4 mL); 1 fluid ounce equals 29.57 mL (2957 cm³).

Density The **density (d)** of an object is its mass divided by its volume:

$$\text{Density} = \text{mass/volume}$$

We can also rearrange this equation and solve for mass or volume:

$$\text{Mass} = \text{volume} \times \text{density}$$

$$\text{or Volume} = \text{mass/density}$$

Because volume can change with temperature, so can density. But, at a given temperature and pressure, the *density of a substance is a characteristic physical property and, thus, has a specific value.*

The SI unit of density is **kilograms per cubic meter (kg/m³)**, but in chemistry, density has units of g/L (g/dm³) or g/mL (g/cm³). Table 1.4 shows densities of some common substances. Note that the densities of gases are much lower than those of liquids or solids.

Substance	Physical State	Density (g/cm ³)
Hydrogen	gas	0.0000899
Oxygen	gas	0.00133
Grain alcohol	liquid	0.789
Water	liquid	0.998
Table salt	solid	2.16
Aluminum	solid	2.70
Lead	solid	11.3
Gold	solid	19.3

* At room temperature (20°C) and normal atmospheric pressure (1 atm).

Significant Figures in Measurements

Why use significant figures?

Significant figures are the digits in a number that carry meaningful information about its precision. They include all the certain digits plus the first uncertain (or estimated) digit in a measurement. Significant figures indicate the precision of a measurement or calculation. Using the correct number of significant figures helps ensure that reported results are neither over- nor under-stated in terms of accuracy. They help communicate the reliability of measurements and results. For example, if you measure something as 12.3 cm, it implies precision to the nearest tenth of a centimeter, not to the nearest millimeter.

Rules for Identifying Significant Figures:

1. All non-zero digits ARE significant.

Example: 123.45 has 5 significant figures.

2. Zeros between two non-zero digits ARE significant.

Example: 1002 has 4 significant figures; 2025 has 4 significant numbers; 309 has 3 significant numbers)

3. Leading zeros (zeros before the first non-zero digit) are NOT significant.

Example: 0.0045 has 2 significant figures; 0.54 has only TWO significant figures.

4. Trailing zeros to the right of the decimal ARE significant.

Example: 45.00 has 4 significant figures; 0.0350 (3 significant numbers)

45.00 is different from 45: a scientist who measures 45.00 grams knows their value to the nearest 1/100th gram; meanwhile their colleague who measured 92 grams only knows their value to the nearest 1 gram. It's important to understand that "zero" does not mean "nothing." Zero denotes actual information, just like any other number.

5. Trailing zeros in a whole number without a decimal point are NOT significant.

Example: Writing 540 indicates that the zero is NOT significant, and there are only TWO significant figures in this value.

Scientific notation can be used to change the number of significant numbers in a value. For example, let's try writing 1500 with THREE significant figures. 1500 has only 2 significant figures since the trailing zeros are not significant, If we add a decimal at the end, 1500., this means we now have FOUR significant figures. But by using scientific notation: 1.50×10^3 , we create a THREE significant figure value.

For example, a length measurement of 94,500 m can be expressed with 3, 4 or 5 significant figures based on how the measurement was obtained:

3 significant figures: 9.45×10^4 m

4 significant figures: 9.450×10^4 m

5 significant figures: 9.4500×10^4 m

6. Exact numbers have an INFINITE number of significant numbers. This rule applies to numbers that are definitions. For example, 1 meter = 100 cm; 1 hour = 60 minutes; 1 in = 2.54 cm.

Rules for rounding off:

1. If the digit to be removed is 5 or above, the preceding number increased by 1 (so round up).
2. If the digit to be removed is less than 5, the preceding number remains the same (so drop without changing).
3. When doing a multistep calculation, always carry an extra one or 2 digits and round off at the end.

Significant figures in Calculations

a) Multiplication and Division

· The final answer should have the **same number of significant figures as the measurement with the fewest number of significant figures**.

Example: $10.5 \text{ cm} \times 2.33 \text{ cm} \times 0.70 \text{ cm} = 17.1255 \text{ cm}^3 = 17 \text{ cm}^3$

The measurement 0.70 has only 2 significant figures, so the final answer must only contain 2 significant figures.

Example: $9.235 \text{ g} / 1.55 \text{ cm}^3 = 5.958 \text{ g/cm}^3 = 5.96 \text{ g/cm}^3$

The measurement 1.55 has 3 significant figures, so the final answer must contain 3 significant figures, before we drop the next digit, we round up.

b) Addition and Subtraction

· The final answer should have the **same number of digits after the decimal point as the measurement with the fewest number of digits after the decimal point**.

Example: $7.55 \text{ g} + 1.2 \text{ g} + 0.875 \text{ g} = 9.625 \text{ g} = 9.6 \text{ g}$

The measurement 1.2 contains only 1 digit after the decimal point, so the final answer must only contain 1 digit after the decimal point.

Example: $16.125 \text{ mL} - 5.\underline{95} \text{ mL} = 10.\underline{175} \text{ mL} = \mathbf{10.18 \text{ mL}}$

The measurement 5.95 contains 2 digits after the decimal point, so the final answer must only contain 2 digits after the decimal point.

c) Mixed Operations

- Do each step in order, keeping track of sig figs at each stage.
- Apply rounding at the **final step only**, not in intermediate steps (to avoid rounding error). It is recommended to carry an extra digit or 2 until the end.

Example:

Calculate the density of a cube that has a side length of 1.55 cm, and has a mass of 955 g.

Answer

Rounding after each step:

First we must calculate the volume = $1.55 \text{ cm} \times 1.55 \text{ cm} \times 1.55 \text{ cm} = 3.72 \text{ cm}^3$

Density = mass/volume = $955 \text{ g} / 3.72 \text{ cm}^3 = \mathbf{257 \text{ g/cm}^3}$

Rounding only at the final step:

First we must calculate the volume = $1.55 \text{ cm} \times 1.55 \text{ cm} \times 1.55 \text{ cm} = 3.724 \text{ cm}^3$ (we will carry an extra digit until the end)

Density = mass/volume = $955 \text{ g} / 3.724 \text{ cm}^3 = \mathbf{256 \text{ g/cm}^3}$

*Always remember to carry an extra digit or 2 and round at the final step only!

Unit Conversions and Dimensional Analysis

In chemistry and science, **measurements are only meaningful when they include units**. Often we need to convert a measurement from one unit to another—for example meters to centimeters, grams to kilograms, or seconds to hours.

Unit conversions ensure that calculations are consistent and that quantities are expressed in useful forms.

A **conversion factor** is a ratio of two equivalent quantities. Because the numerator and denominator represent the same quantity, the ratio equals **1**, so multiplying by it does not change the value of the measurement.

Example: $1 \text{ m} = 100 \text{ cm}$

Two conversion factors can be written:

$$\frac{1 \text{ m}}{100 \text{ cm}} \quad \text{or} \quad \frac{100 \text{ cm}}{1 \text{ m}}$$

We choose the one that **cancels the unwanted unit**.

Dimensional Analysis

Dimensional analysis is a method of solving conversion problems by treating units as algebraic quantities. We use conversion units to convert between different units. Units can **multiply, divide, and cancel** just like numbers.

Steps in Dimensional Analysis:

1. Write the **given quantity**
2. Multiply by the **appropriate conversion factor**
3. Arrange units so the **unwanted unit cancels**
4. Perform the calculation

Examples:

1. Convert **2.50 m to cm**:

The conversion factor we need to use: $1 \text{ m} = 100 \text{ cm}$

$$2.50 \text{ m} \times \frac{100 \text{ cm}}{1 \text{ m}} = 2.50 \times 10^2 \text{ cm}$$

2. Convert **65 km/h to m/s**:

This is an example where more than one conversion factor is needed, it is an example of a multi-step calculations. The conversion factors we need to use:

$1 \text{ km} = 1000 \text{ m}$ and $1 \text{ h} = 3600 \text{ s}$ (you can also use $1 \text{ h} = 60 \text{ min}$ and $1 \text{ min} = 60 \text{ s}$)

$$65 \frac{\text{km}}{\text{h}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = 18 \text{ m/s}$$

3. Convert **13.6 g/cm³ to lb/in³**:

The conversion factors we need to use: $1 \text{ lb} = 454 \text{ g}$ and $1 \text{ in} = 2.54 \text{ cm}$

$$13.6 \frac{\text{g}}{\text{cm}^3} \times \frac{1 \text{ lb}}{454 \text{ g}} \times \frac{(2.54 \text{ cm})^3}{(1 \text{ in})^3} = 0.491 \text{ lb/in}^3$$

Accuracy and Precision

Understanding the difference between **accuracy** and **precision** is essential in scientific measurements because it helps us evaluate the **quality and reliability of experimental data**.

Accuracy

Accuracy refers to how **close a measured value is to the true or accepted value** of a quantity.

- A measurement is **accurate** if it is very close to the correct value.
- Accuracy is affected mainly by **systematic errors**, such as faulty equipment or incorrect calibration.

Example

Accepted value of density of water at 25°C is **0.997 g/mL**

Measurement	Accuracy
0.996 g/mL	Very accurate
0.998 g/mL	Very accurate
1.30 g/mL	Not accurate

If your measurement is close to the true value, it is **accurate**.

Precision

Precision refers to how **close repeated measurements are to each other**.

- High precision means **consistent or reproducible results**.
- Precision is mainly affected by **random errors** in measurement.

Example

Measurement Set	Precision
0.996, 0.997, 0.998	High precision
0.70, 1.20, 1.50	Low precision

If measurements are **very similar to each other**, they are precise.

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1.2: Student Activity

Student Activity: In groups of 3-4, answer each of the following question. Make sure you show ALL your work for full credit.

- How long (in minutes) will it take to travel the 406 miles between Los Angeles and San Francisco at an average speed of 48 miles per hour.
- Make each conversion indicated below. Use scientific notation to avoid long integers or decimal fractions.
 - 15.3 kg to g and mg
 - 0.666 km to cm
 - 90.8 mL to L to cm^3
- Round off each quantity to three significant figures:
 - 6.398×10^{-3} km rope
 - 79,000 m cable
 - 42,150 tons fertilizer
 - \$649.85
 - 52.20 mL helium
 - 23,642,000 mm wavelength
 - 0.0041963 kg lead
- The height of Angel Falls in Venezuela is 979 m. How high is this in (a) yards; (b) feet?
- Fill in the spaces in the following table so that each temperature is expressed in all three scales:

Celsius	Fahrenheit	Kelvin
68		
	-29	
		111
	36	
		358
-141		

- A rectangular block of iron with dimensions 4.60 cm, 10.3 cm, and 13.2 cm, has a mass of 4.92 kg. Find its density in g/cm^3 . Make sure to use the correct number of significant figures.

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1.3: Assessment

Instructions: You will be given 30 minutes to complete this quiz. Please answer each question to the best of your ability using the correct number of significant figures and appropriate units when necessary. Proper work must be shown to receive full credit.

1. In determining the density of a rectangular metal bar, a student made the following measurements: length = 3.30 cm; width = 1.4 cm; height = 0.80 cm; and the mass = 18.4800 g. Calculate the density of the metal to the correct number of significant figures. **[4 points]**
2. Perform the following conversions: **[4 points]**:
 - a. The diameter of Earth is 12.7 Mm. Express this diameter in centimeters.
 - b. A gas can holds 5.0 gallons of gasoline. Express this quantity in cm^3 .
3. The density of iron is 7.86 g/cm^3 . What is its density in pounds per cubic inch (lb/in^3). **[4 points]**
4. The warmest temperature ever measured in the United States is 134°F on July 10, 1913, in Death Valley, California. Convert that temperature to $^\circ\text{C}$ and K. **[3 points]**
5. Carry out the following operations as if it were a calculation of real experimental results. Provide units where applicable and use the appropriate number of significant figures. Express the answers in scientific notation, if needed. **[6 points]**
 - a. $[(1.7 \times 106) \div (2.63 \times 105)] + 7.33 =$
 - b. $(85.3 - 21.489) \div 0.0059 =$
 - c. $(24.6681 \times 2.38) + 332.58 =$

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CHAPTER OVERVIEW

2: Percent Composition and Empirical and Molecular Formulas

Introduction to Module

Chemists often need to determine the composition of compounds and understand the relationships between the elements that make them up. Three important tools used for this purpose are **percent composition by mass**, **empirical formulas**, and **molecular formulas**. These concepts help scientists determine how much of each element is present in a compound and how those elements combine to form chemical substances.

Percent composition by mass describes the percentage of the total mass of a compound that comes from each element in the compound. It provides useful information about the relative amounts of elements present and can be calculated from a compound's chemical formula or determined experimentally from laboratory data. Percent composition is widely used in analytical chemistry, quality control, and determining unknown substances.

From percent composition data, chemists can determine a compound's **empirical formula**, which represents the **simplest whole-number ratio of atoms** of each element in the compound. The empirical formula does not necessarily show the actual number of atoms in a molecule but instead reflects the basic proportional relationship among the elements.

The **molecular formula**, on the other hand, shows the **actual number of atoms of each element in a molecule** of a compound. It is usually determined by comparing the empirical formula mass with the compound's molar mass. The molecular formula is always a whole-number multiple of the empirical formula.

One common experimental technique used to determine empirical formulas, especially for organic compounds, is **combustion analysis**. In this method, a compound is burned in excess oxygen, producing carbon dioxide and water. By measuring the masses of these products, chemists can determine the amounts of carbon and hydrogen originally present in the sample. From this information, along with the mass of the original compound, the empirical formula can be calculated.

Together, percent composition, empirical and molecular formulas, and combustion analysis provide powerful methods for identifying chemical compounds and understanding their composition. These techniques form an important part of quantitative chemical analysis and help connect experimental measurements with the formulas used to represent chemical substances.

Learning Outcomes

By the end of this module, students are expected to:

- Compute the percent composition of a compound.
- Determine the empirical formula of a compound.
- Determine the molecular formula of a compound.
- Determine the empirical formula of a compound from combustion analysis.

[2.1: Percent Composition and Empirical and Molecular Formulas](#)

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2.1: Percent Composition and Empirical and Molecular Formulas

Percent Composition of Compounds

The subscripts in the chemical formula of a compound indicate how many atoms of each element are present in the compound. For example, the chemical formula for H₂O indicates that there are 2 H atoms and 1 O atom in a molecule of water. Using the chemical formula, we can calculate the percent of each element in the compound. This is called the **percent composition by mass**.

The percent composition is the percentage by mass of each element in a chemical formula. This concept is a direct application of the law of constant composition, which states that any pure sample of a compound will always contain the same elements in the same proportion by mass, regardless of the sample/s size or origin.

The percent composition can be calculated using the following formula:

$$\text{Mass \% of element } X = \frac{n \times \text{molar mass of } X}{\text{molar mass of compound}} \times 100\%$$

where n is the number of atoms of the element in the compound.

Solved Example

Tooth enamel is Ca₅(PO₄)₃(OH). Calculate the percent composition of the elements present.

Step 1. Calculate the molar mass of the compound:

$$\text{molar mass} = [(5 \times 40.08) + (3 \times 30.97) + (1 \times 1.008) + (13 \times 16.00)] = 502.32 \frac{\text{g}}{\text{mol}}$$

Step 2. Calculate the mass % of each element:

$$\% \text{ Ca} = \frac{5 \times 40.08}{502.32} \times 100\% = 39.89\%$$

$$\% \text{ P} = \frac{3 \times 30.97}{502.32} \times 100\% = 18.50\%$$

$$\% \text{ O} = \frac{13 \times 16.00}{502.32} \times 100\% = 41.41\%$$

$$\% \text{ H} = 1 \times \frac{1.008}{502.32} \times 100\% = 0.2007\%$$

Practice 1.

Lithium carbonate, Li₂CO₃, was the first “mood-stabilizing” drug approved by the FDA for the treatment of mania and manic-depressive illness, also known as bipolar disorder. Calculate the percent composition by mass of lithium carbonate.

Practice 2.

Cisplatin, or Platinol, is used in the treatment of certain cancers. The chemical formula of this compound is Pt(NH₃)₂Cl₂. Calculate the percent composition of cisplatin.

Empirical Formula of a Compound

The **empirical formula** provides the **simplest whole-number ratio** of elements in a compound. It is determined by experimental data (such as combustion analysis) and represents the fundamental building block ratio. The empirical formula tells us the relative number of atoms of each element in the compound. The empirical formula for CH₄ shows that this compound contains one C atom and four hydrogen atoms. This ratio also applies on the molar level: 1 mol of CH₄ contains 1 mol of C and 4 moles of H. Hence, *the ratio of the numbers of moles of all elements in a compound gives the subscripts in the compound's empirical formula*. Thus, the mole concept provides a way of calculating the empirical formula.

To find the empirical formula from mass percent, assume a 100-gram sample to convert percentages to grams. (Samples of any size can be used in these types of problems; however, we generally use 100.0 g-sample to simplify the calculation of mass from percentage: % VALUE = MASS VALUE IN GRAMS). Using the atomic masses of elements from the periodic table to get molar

masses, we convert grams to moles using molar mass (**we calculate the number of moles of each element in the sample**). We then divide all mole values by the smallest number of moles to obtain the molar ratio of the elements in the formula. If the resulting numbers are not whole, multiply them by the smallest integer that will turn them into whole numbers to find the subscripts for the empirical formula.

Solved Example

Determine the empirical formula for a compound with 69.94% iron (Fe) and 30.06% oxygen (O)

1. Convert percentages to grams: Assume a 100-gram sample, so you have 69.94 g of Fe and 30.06 g of O.

2. Convert grams to moles: Use the molar masses from the periodic table (Fe = 55.85 g/mol and O is 16.00 g/mol) to convert the grams to moles.

$$69.94 \text{ g Fe} \times \frac{1 \text{ mol Fe}}{55.85 \text{ g Fe}} = 1.252 \text{ mol Fe}$$

$$30.06 \text{ g O} \times \frac{1 \text{ mol O}}{16.00 \text{ g O}} = 1.879 \text{ mol O}$$

3. Find the smallest number of moles: The smallest value is 1.252 mol of Fe.

4. Divide all mole values by the smallest value:

$$\frac{1.252 \text{ mol Fe}}{1.252} = 1 \text{ Fe}$$

$$\frac{1.879 \text{ mol O}}{1.252} = 1.5 \text{ O}$$

5. Convert to whole numbers: Since you have a decimal of .5, multiply all the resulting numbers by 2 to get whole numbers.

$$1 \text{ Fe} \times 2 = 2 \text{ Fe}$$

$$1.5 \text{ O} \times 2 = 3 \text{ O}$$

6. Write the empirical formula: The empirical formula is: Fe₂O₃.

General Procedure:

Step 1 — Convert % values to grams: % value = mass value

Step 2 — Convert to moles: divide each by its molar mass

Step 3 — Divide ALL mole values by smallest number of moles to determine the whole-number ratio of moles

Step 4 — Multiply by smallest integer to convert to whole numbers, if needed (i.e., If you get decimals like 1.5 or 2.33 → multiply all by 2, 3, or 4 to make whole numbers.)

Solved Example

Methyl benzoate, a compound used in the manufacture of perfumes, contains 70.57 % C, 5.934 % H, and 23.49 % O by mass. What is the empirical formula of this compound?

Solution:

(1) Assume 100-g sample: % value = mass value in grams

$$70.57\% \text{ C} = 70.57 \text{ g C} \quad 5.934\% \text{ H} = 5.934 \text{ g H} \quad 23.49\% \text{ O} = 23.49 \text{ g O}$$

(2) Find the moles of each element using its molar mass:

$$70.57 \text{ g C} \times \frac{1 \text{ mol C}}{12.01 \text{ g C}} = 5.876 \text{ mol C}$$

$$5.934 \text{ g H} \times \frac{1 \text{ mol H}}{1.008 \text{ g H}} = 5.887 \text{ mol H}$$

$$23.49 \text{ g O} \times \frac{1 \text{ mol O}}{16.00 \text{ g O}} = 1.468 \text{ mol O}$$

(3) Divide all by the smallest number of moles to determine the whole-number ratio of moles:

$$\frac{5.876 \text{ mol } C}{1.468} = 4 \text{ } C \quad \frac{5.887 \text{ mol } H}{1.468} = 4 \text{ } H \quad \frac{1.468 \text{ mol } O}{1.468} = 1 \text{ } O$$

Empirical formula = C₄H₄O

Practice Exercise 1

Ascorbic acid (Vitamin C) contains 40.92% C, 4.58% H, and 54.50% O. What is the empirical formula of ascorbic acid?

Practice Exercise 2

Phosgene, a compound used as a chemical warfare during World War I, contains 0.260 g of carbon, 0.347 g of oxygen, and 1.537 g of chlorine. What is the empirical formula of this substance?

Molecular Formula from Empirical Formula

The **molecular formula** provides the **exact number** of atoms of each element in a compound. We can obtain the molecular formula for any compound from its empirical formula if we know its molecular mass or the molar mass. The subscripts in the molecular formula are whole-number multiples of the subscripts in its empirical formula.

Examples:

· Glucose has molecular formula **C₆H₁₂O₆**

→ Empirical formula = **CH₂O**

· Hydrogen peroxide: **H₂O₂**

→ Empirical formula = **HO**

This multiple can be found by dividing the molar mass by the mass of the empirical formula:

$$\text{whole - number multiple} = \frac{\text{molar mass}}{\text{mass of empirical formula}}$$

Solved Example

A compound has empirical formula **CH₂O** and molar mass **180 g/mol**. Determine the molecular formula of the compound.

Step 1 — Calculate empirical formula mass

Mass of empirical formula = [(1 × 12.01) + (2 × 1.008) + (1 × 16.00)] = 30.03 g/mol

Step 2 — Find whole-number multiple

$$\text{Whole - number multiple} = \frac{\text{molar mass}}{\text{mass of empirical formula}}$$

$$\text{Whole - number multiple} = \frac{180}{30.03} = 6$$

Step 3 — Multiply empirical formula by the whole-number multiple

CH₂O × 6 → C₆H₁₂O₆

Molecular formula = C₆H₁₂O₆

More Practice:

Example 1 — From Percent Composition

Ethylene glycol, used in automobile antifreeze, is 38.7% C, 9.7% H, and 51.6% O by mass. Its molar mass is 62.1 g/mol. What is the empirical formula and molecular formula of ethylene glycol.

Solution

Assume 100 g sample; % value = mass value:

$$38.7\% \text{ } C = 38.7 \text{ g } C \quad 9.7\% \text{ } H = 9.7 \text{ g } H \quad 51.6\% \text{ } O = 51.6 \text{ g } O$$

Convert mass to moles:

$$38.7 \text{ g C} \times \frac{1 \text{ mol C}}{12.01 \text{ g C}} = 3.22 \text{ mol C} \quad 9.7 \text{ g H} \times \frac{1 \text{ mol H}}{1.008 \text{ g H}} = 9.6 \text{ mol H} \quad 51.6 \text{ g O} \times \frac{1 \text{ mol O}}{16.00 \text{ g O}} = 3.23 \text{ mol O}$$

Divide ALL by smallest number of moles (3.22):

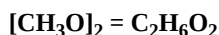
$$\frac{3.22 \text{ mol C}}{3.22} = 1 \text{ C} \quad \frac{9.6 \text{ mol H}}{3.22} = 3 \text{ H} \quad \frac{3.23 \text{ mol O}}{3.22} = 1 \text{ O}$$

Empirical Formula: CH_3O

Next, we must find the whole-number multiple:

$$\text{whole - number multiple} = \frac{62.1}{31.03} = 2$$

Molecular formula:



Practice 1:

Cyclohexane, a commonly used organic solvent, is 85.6% C and 14.4% H by mass with a molar mass of 84.2 g/mol. What is its molecular formula?

Practice 2:

Caffeine, a stimulant found in coffee, contains 49.5% C, 5.15% H, 28.9% N, and 16.5% O by mass. Its molar mass is 195 g/mol. Determine its molecular formula.

Empirical Formula from Combustion Analysis

Combustion analysis involves burning a known mass of a substance (often an organic compound containing C and H, and possibly O, N, or S) in excess oxygen and measuring the masses of the products, typically carbon dioxide (CO_2) and water (H_2O). Combustion analysis is used in the laboratory to determine the empirical formulas of compounds containing mainly carbon and hydrogen, and sometimes oxygen.

To determine the **empirical formula** of a compound from combustion analysis, using an apparatus such as that shown in Figure 3.1, you first find the mass of each element (typically carbon and hydrogen, sometimes oxygen or others) in the sample by analyzing the combustion products (CO_2 and H_2O). The masses of these compounds are determined experimentally. Using the mass of CO_2 and the mass of H_2O , we can calculate the number of moles of C and H, respectively, in the original sample. We then convert these masses to moles, find the simplest mole ratio, and use this ratio as subscripts for the empirical formula. If a third element is present, we can determine its mass by subtracting the sum of masses of C and H from the mass of the original sample. Once we know the mass of each element, we can determine the empirical formula following the method we established earlier.

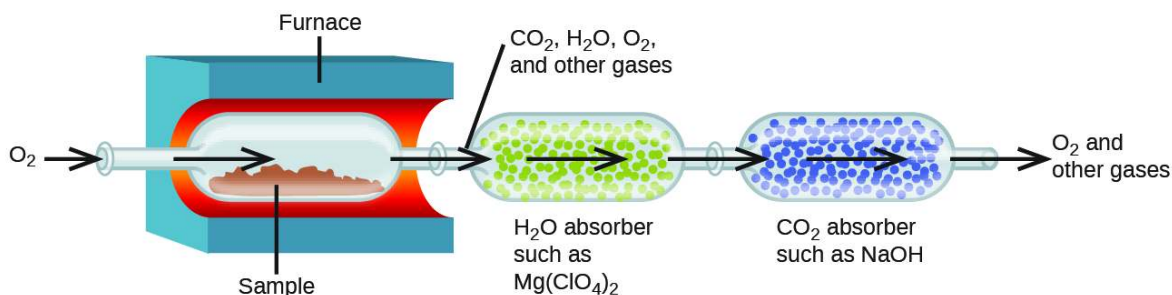


Figure 4.1. This schematic diagram illustrates the basic components of a combustion analysis device for determining the carbon and hydrogen content of a sample.

(https://chem.libretexts.org/Bookshel...mical_Analysis)

Solved Example:

A 0.255 g sample of isopropyl alcohol (which contains only C, H, and O) is burned in a combustion apparatus. The combustion reaction produces 0.561 g of CO₂ and 0.306 g of H₂O. What is the empirical formula for isopropyl alcohol?

Step 1: Calculate the Mass of Carbon from the mass of CO₂ and the mass of Hydrogen from the mass of H₂O:

· **Mass of C:** All the carbon in the original sample is converted to C in the CO₂. Use the mass of CO₂ produced and the molar mass ratio to find the mass of C.

$$0.561 \text{ g CO}_2 \times \frac{1 \text{ mol CO}_2}{44.01 \text{ g CO}_2} \times \frac{1 \text{ mol C}}{1 \text{ mol CO}_2} \times \frac{12.01 \text{ g C}}{1 \text{ mol C}} = 0.153 \text{ g C}$$

· **Mass of H:** All the hydrogen in the sample is converted to H in the H₂O. Use the mass of H₂O produced and the molar mass ratio to find the mass of H.

$$0.306 \text{ g H}_2\text{O} \times \frac{1 \text{ mol H}_2\text{O}}{18.02 \text{ g H}_2\text{O}} \times \frac{2 \text{ mol H}}{1 \text{ mol H}_2\text{O}} \times \frac{1.008 \text{ g H}}{1 \text{ mol H}} = 0.0342 \text{ g H}$$

Step 2: Calculate the mass of oxygen

Mass of O: Since the compound contains C, H and O only, the mass of O is found by subtracting the total mass of C and H calculates above from the mass of the original sample.

$$\text{mass of O} = 0.255 \text{ g sample} - [0.153 \text{ g C} + 0.0342 \text{ g H}] = 0.068 \text{ g O}$$

Step 3: Convert Masses to Moles

Convert the mass of each element to moles using its atomic mass.

$$0.153 \text{ g C} \times \left(\frac{1 \text{ mol C}}{12.01 \text{ g C}} \right) = 0.0128 \text{ mol C}$$

$$0.0342 \text{ g H} \times \left(\frac{1 \text{ mol H}}{1.008 \text{ g H}} \right) = 0.0339 \text{ mol H}$$

$$0.068 \text{ g O} \times \left(\frac{1 \text{ mol O}}{16.00 \text{ g O}} \right) = 0.0043 \text{ mol O}$$

Step 4: Divide all by the smallest number of moles to determine the whole-number ratio of moles:

$$\text{C: } \frac{0.0128 \text{ mol C}}{0.0043} = 3.0 \quad \text{H: } \frac{0.0339 \text{ mol H}}{0.0043} = 7.9 \quad \text{O: } \frac{0.0043 \text{ mol O}}{0.0043} = 1.0$$

Empirical formula of isopropyl alcohol is: C₃H₈O.

Practice Exercise 1

Combustion analysis of toluene, a common organic solvent, gives 5.86 mg of CO₂ and 1.37 mg of H₂O. If the compound contains only carbon and hydrogen, what is its empirical formula?

Practice Exercise 2

Menthol, the substance we can smell in mentholated cough drops, is composed of C, H, and O. A 0.1005-g sample of menthol is combusted, producing 0.2829 g of CO₂ and 0.1159 g of H₂O. What is the empirical formula of menthol? If menthol has a molar mass of 156 g/mol, what is its molecular formula?

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2.2: Student Activity

In groups of 3-4, answer each of the following questions. Show ALL your work clearly for full credit.

1. Calculate the percentage composition by mass of aluminum sulfate.
2. How many grams of fluorine are in 216 g of calcium fluoride?
3. A sample of a pure compound is made up of 1.61 g of phosphorus and 2.98 g of fluorine. Find the empirical formula of the compound.
4. The mass of a piece of iron is 1.62 g. If the iron is exposed to oxygen under conditions in which oxygen combines with all of the iron to form pure oxide of iron, and the final mass increases to 2.31 g. Find the empirical formula of the compound.
5. Nicotine is 74.1% carbon, 8.64% hydrogen, and 17.3% oxygen. What is the empirical formula of nicotine?
6. An antifreeze and coolant widely used in automobile engines is 38.7% carbon, 9.7% hydrogen, and 51.6% oxygen. Its molar mass is 62.0 g/mol. What is the molecular formula of the compound?
7. Tartaric acid is the white, powdery substance that coats tart candies such as Sour Patch Kids. Combustion analysis of a 12.01 g sample of tartaric acid-which contains only carbon, hydrogen, and oxygen- produced 14.08 g CO_2 and 4.32 g H_2O . Determine the empirical formula for tartaric acid.

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2.3: Assessment

Instructions: Please answer each question to the best of your ability using the correct number of significant figures and appropriate units when needed.

1. Calculate the % by mass of Boron (B) in $\text{Ca}(\text{BO}_2)_2 \cdot 6\text{H}_2\text{O}$. **[3 points]**
2. What is the mass percent of sodium in sodium carbonate? **[3 points]**
3. Determine the empirical formula of a compound that contains 43.64 g of P and 56.36 g of O in a particular sample. What is the name of the compound? **[4 points]**
4. What is the empirical formula of a substance that contains 1.55 g of carbon, 0.260 g of hydrogen, and 3.096 g of oxygen? If the molar mass was found to be 152.10 g/mol, determine the molecular formula? SHOW YOUR WORK. **[5 points]**
5. Terephthalic acid, used in the production of polyester fibers and films, is composed of carbon, hydrogen, and oxygen. When 0.6943 g of terephthalic acid was subjected to combustion analysis, it produced 1.471 g CO_2 and 0.226 g H_2O . What is the empirical formula of the compound? **[5 points]**

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CHAPTER OVERVIEW

3: Ionic and Covalent Bonding

Introduction to Module

Chemical compounds form when atoms combine through chemical bonds to achieve more stable electron arrangements. Two of the most common types of compounds are **ionic compounds** and **covalent compounds**, which differ in how the atoms are held together and how they are represented in chemical formulas.

Ionic compounds form when electrons are transferred from one atom to another, usually between a metal and a nonmetal. When a metal atom loses one or more electrons, it becomes a **positively charged ion (cation)**, while the nonmetal that gains electrons becomes a **negatively charged ion (anion)**. The opposite charges attract, forming an ionic bond and creating a compound arranged in a crystal lattice. Examples include sodium chloride (NaCl) and calcium chloride (CaCl₂).

Covalent compounds, also called **molecular compounds**, form when atoms—typically nonmetals—share electrons to achieve stable electron configurations. Instead of forming ions, the atoms remain neutral and are held together by shared pairs of electrons. Covalent compounds form discrete molecules such as water (H₂O), carbon dioxide (CO₂), and methane (CH₄).

Because many compounds exist, chemists use systematic rules to **name compounds** so that each substance has a unique and universally understood name. **Ionic compounds** are named by listing the cation first followed by the anion. For metals that can form more than one ion, **Roman numerals** indicate the metal's charge. **Covalent compounds**, on the other hand, use **Greek prefixes** (such as mono-, di-, tri-, tetra-) to indicate the number of atoms of each element in a molecule.

Understanding the differences between ionic and covalent compounds and learning the rules for naming them are essential skills in chemistry. These naming systems allow chemists to clearly communicate the composition of substances and interpret chemical formulas used throughout chemical science.

Learning Outcomes

By the end of this module, students are expected to:

- Explain the essential features of ionic and covalent compounds and distinguish between them
- Predict the monatomic ion formed from a main-group element
- Name and write the formula of binary ionic and covalent ionic compounds
- Name and write the formula of ionic compounds that contain polyatomic ions
- Name and write the formula of acids and hydrates

[3.1: Introduction to Ionic and Covalent Bonding](#)

[3.2: Student Activity](#)

[3.3: Assessment](#)

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3.1: Introduction to Ionic and Covalent Bonding

Chemical bonding lies at the heart of chemistry. Nearly everything you see—water, salt, proteins, metals, minerals—is held together by forces that arise when atoms interact. In this chapter, we explore the two major types of chemical bonds: **ionic bonds** and **covalent bonds**. Understanding these allows us to explain why substances have the properties they do, how compounds form, and how atoms achieve stability.

Why Atoms Form Bonds

Atoms tend to bond in order to achieve a more stable electron arrangement, often resembling the **noble gas electron configuration**. This stability is usually achieved by either **transferring** electrons or **sharing** them.

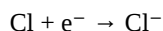
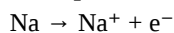
- When electrons are **transferred**, an ionic bond forms.
- When electrons are **shared**, a covalent bond forms.

Formation of Ions

Ionic bonding occurs when an atom with low ionization energy (usually a metal) transfers electrons to an atom with high electron affinity (usually a nonmetal).

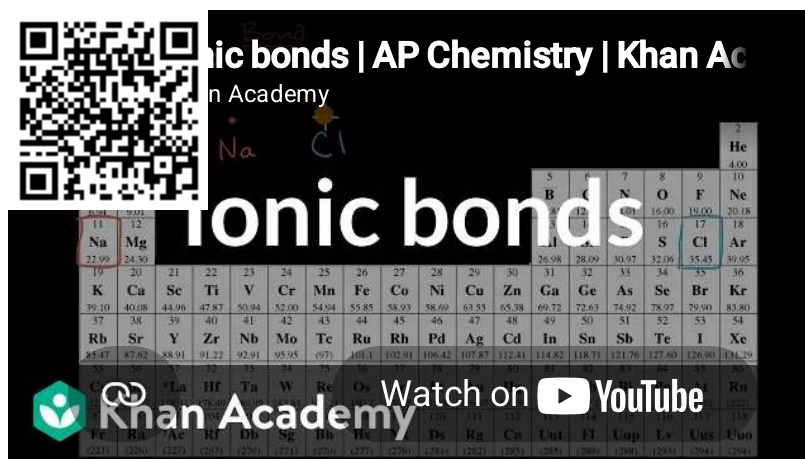
- A metal **loses electrons** → becomes a **cation** (positively charged ion)
- A nonmetal **gains electrons** → becomes an **anion** (negatively charged ion)

Example:



Na^+ and Cl^- attract to form NaCl.

Ionic bonding typically occurs between **metals and nonmetals**. When metals lose electrons, they form positively charged ions (**cations**), while nonmetals gain electrons to become negatively charged ions (**anions**). The **electrostatic attraction** between these oppositely charged ions forms a strong ionic bond.



Thumbnail for a Khan Academy video titled "Ionic bonds | AP Chemistry | Khan Academy". The image shows a periodic table with Sodium (Na) and Chlorine (Cl) highlighted. The text "Ionic bonds" is prominently displayed in the center. A QR code is visible in the top left corner, and the Khan Academy logo and "Watch on YouTube" text are at the bottom.

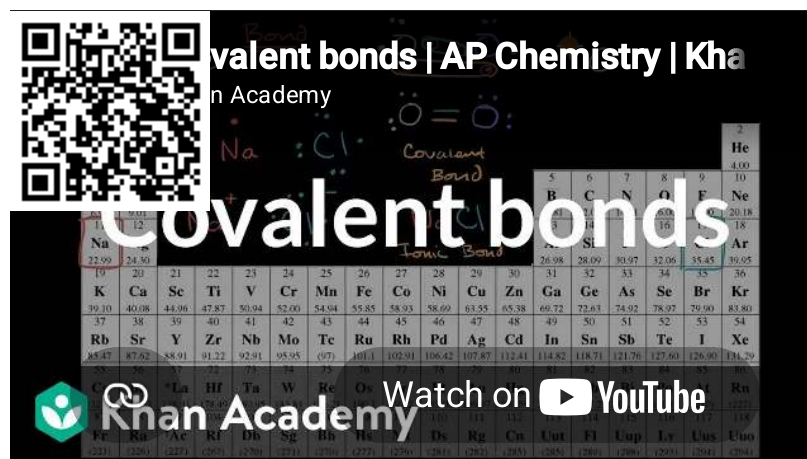
Note: All Khan Academy content is available for free at (www.khanacademy.org).

Key features of ionic compounds:

- Form crystal lattice structures
- Usually have high melting and boiling points
- Conduct electricity when molten or dissolved in water
- Often dissolve in water

A familiar example is **sodium chloride (NaCl)**—table salt.

Covalent bonding usually occurs between **two nonmetals**. Instead of transferring electrons, the atoms **share electron pairs** in order to achieve a stable noble gas configuration. A shared pair of electrons constitutes a covalent bond.



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Key features of covalent compounds:

- Can form single, double, or triple bonds
- Often exist as molecules (like H₂O or CO₂)
- Have lower melting and boiling points compared to ionic compounds
- Often do not conduct electricity

Binary ionic compounds are composed of **two elements**:

1. A **metal cation** (positive ion)
2. A **nonmetal anion** (negative ion)

General Rules for Naming Binary Ionic Compounds

1. Name the cation first.

- The cation name is the same as the element name.

Na⁺ → *sodium ion*

Ca²⁺ → *calcium ion*

Metals with a fixed charge

Some metals form **only one possible charge**. These metals DO NOT need a Roman numeral when naming them.

Always have one charge:

- Group 1 metals: +1
- Group 2 metals: +2
- Aluminum (Al³⁺), Zinc (Zn²⁺), Silver (Ag⁺), Cadmium (Cd²⁺)

Metals with variable charge

Many transition metals can form cations of **multiple possible charges**. For these metals, we **MUST indicate the metal's charge using a Roman numeral** in parentheses when naming these ions.

Examples:

- Fe²⁺: Iron(II) ion Fe³⁺: Iron(III) ion
- Cu⁺: copper(I) ion Cu²⁺: copper(II) ion

- Sn^{2+} : tin(II) ion Sn^{4+} : tin(IV) ion
- Pb^{2+} : lead(II) ion Pb^{4+} : lead(IV) ion

2. Name the anion second.

- Use the root of the element's name and change the ending to **-ide**.

Cl^- → chloride ion O^{2-} → oxide ion

S^{2-} → sulfide ion N^{3-} → nitride ion

Br^- → bromide ion P^{3-} → phosphide ion

3. Name of the compound = Cation Name + Anion Name

Examples:

- NaCl → sodium chloride
- CaO → calcium oxide
- AgBr → silver bromide
- FeO → iron(II) oxide
- CrI_3 → chromium(III) iodide

How to determine the Roman numeral:

Use the anion charge to figure out the metal's charge.

Examples (with reasoning):

FeCl_2

- Cl^- has a -1 charge
- $2 \text{Cl}^- = -2$
- Fe must be $+2$
→ **iron(II) chloride**

FeCl_3

- $3 \text{Cl}^- = -3$
- Fe must be $+3$
→ **iron(III) chloride**

CuO

- $\text{O}^{2-} = -2$
- Cu must be $+2$
→ **copper(II) oxide**

Cu_2O

- $\text{O}^{2-} = -2$

- Two Cu must total +2 → each Cu is +1
→ **copper(I) oxide**

Table 2.1 Common Monatomic Cations

Charge	Cations Formula	Name		Cations Formula	Name
+1	H ⁺	hydrogen		Au ⁺	gold(I)
	Li ⁺	lithium		Cu ⁺	copper(I)
	Na ⁺	sodium			
	K ⁺	potassium			
	Cs ⁺	cesium			
	Ag ⁺	silver			
+2	Mg ²⁺	magnesium		Co ²⁺	cobalt(II)
	Ca ²⁺	calcium		Cu ²⁺	copper(II)
	Sr ²⁺	strontium		Fe ²⁺	iron(II)
	Ba ²⁺	barium		Mn ²⁺	manganese(I)
	Zn ²⁺	zinc		Hg ²⁺	mercury(II)
	Cd ²⁺	cadmium		Ni ²⁺	nickel(II)
				Pb ²⁺	lead(II)
				Sn ²⁺	tin(II)
+3	Al ³⁺	aluminum		Cr ³⁺	chromium(II)
				Fe ³⁺	iron(III)
				Au ³⁺	gold(III)
+4				Pb ⁴⁺	lead(IV)
				Sn ⁴⁺	tin(IV)

Period	Periodic Table of the Elements																	
	1	2											13	14	15	16	17	18
1																		He
2	Li ⁺	Be ²⁺													N ³⁻	O ²⁻	F ⁻	Ne
3	Na ⁺	Mg ²⁺											Al ³⁺		P ³⁻	S ²⁻	Cl ⁻	Ar
4	K ⁺	Ca ²⁺	Sc ⁺³			Cr ³⁺	Mn ⁺³	Fe ²⁺	Co ⁺²	Ni ⁺²	Cu ⁺	Zn ²⁺	Ga ⁺³		As ³⁻	Se ²⁻	Br ⁻	Kr
5	Rb ⁺	Sr ²⁺	Y ⁺³			Cr ⁶⁺	Mn ⁺³	Fe ³⁺	Co ³⁺	Ni ⁺³	Cu ²⁺							
6	Cs ⁺	Ba ²⁺									Au ⁺	Hg ²⁺²⁺					At ⁻	Rn
7	Fr ⁺	Ra ²⁺									Au ³⁺	Hg ²⁺						

Figure : Note some metals form more than one type of monatomic cation (this table is not complete), and Hg_2^{2+} is not really a monatomic ion, but a diatomic ion.

https://chem.libretexts.org/Courses/University_of_Arkansas_Little_Rock/Chem_1300%3A_Preparatory_Chemistry/Learning_Modules/04%3A_Compounds_and_Molecules/4.03%3A_Ionic_Compounds_and_Formulas

A polyatomic ion is a charged species composed of two or more atoms held together by covalent bonds that behave as a single, discrete unit.

Unlike a monatomic ion (a single charged atom like Na^+ or Cl^-), a **polyatomic ion** is a molecule that has gained or lost electrons, resulting in an overall net electrical charge.

Naming ionic compounds that contain polyatomic ions follows the same general principles as naming binary ionic compounds.

Examples:

Cation	Anion	Chemical Formula	Name
Na^+	SO_4^{2-}	Na_2SO_4	sodium sulfate
Ca^{2+}	NO_3^-	$\text{Ca}(\text{NO}_3)_2^*$	calcium nitrate
Fe^{3+}	SO_4^{2-}	$\text{Fe}_2(\text{SO}_4)_3$	Iron(III) sulfate
NH_4^+	S^{2-}	$(\text{NH}_4)_2\text{S}$	Ammonium sulfide

*Parentheses are used in the formula to show how many of the polyatomic ions are present per formula.

Table 2.3 Common Polyatomic Ions

Polyatomic Cations

Formula	Name
NH_4^+	ammonium
H_3O^+	hydronium
Hg_2^{2+}	mercury(I)

Polyatomic Anions

Formula	Name	Formula	Name
CH_3COO^- (or $\text{C}_2\text{H}_3\text{O}_2^-$)	acetate	ClO^-	hypochlorite
CN^-	cyanide	ClO_2^-	chlorite
OH^-	hydroxide	ClO_3^-	chlorate
O_2^{2-}	peroxide	ClO_4^-	perchlorate
CO_3^{2-}	carbonate	NO_2^-	nitrite
HCO_3^-	hydrogen carbonate (or bicarbonate)	NO_3^-	nitrate
CrO_4^{2-}	chromate	MnO_4^-	permanganate
$\text{Cr}_2\text{O}_7^{2-}$	dichromate	PO_4^{3-}	phosphate
SO_3^{2-}	sulfite	HPO_4^{2-}	hydrogen phosphate
SO_4^{2-}	sulfate	H_2PO_4^-	dihydrogen phosphate
HSO_4^-	hydrogen sulfate (or bisulfate)		

Names and Formulas of Acids

Acids are chemical substances that release hydrogen ions (H^+) when dissolved in water. An acid is composed of an anion connected to H^+ ions to neutralize, or balance, the anion's charge.

There are 2 types of acids that we will learn to name. Binary acids are those in which the anion ends with the suffix *-ide* (mainly based on a monatomic anion), and the oxoacids, are those in which the anion ends with either *-ite* or *-ate* (mainly based on a polyatomic anions).

How to name acids:

1. *Acids* containing anions whose names end in *-ide* are named by changing the *-ide* ending to *-ic*, adding the prefix *hydro-* to this anion name, and then following with the word *acid*:

Anion	Corresponding Acid
F^- (fluor <i>ide</i>)	HF (hydrofluor <i>ic</i> acid)
Cl^- (chlor <i>ide</i>)	HCl (hydrochlor <i>ic</i> acid)
CN^- (cyan <i>ide</i>)	HCN (hydrocyan <i>ic</i> acid)

2. Acids containing anions whose names end in **-ate** or **-ite** are named by changing the **-ate** to **-ic** and the **-ite** to **-ous** and then adding the word acid:

Anion	Corresponding Acid
NO_2^- (nitrite)	HNO_2 (nitrous acid)
NO_3^- (nitrate)	HNO_3 (nitric acid)
SO_4^{2-} (sulfate)	• H_2SO_4 (sulfuric acid)
ClO_4^- (perchlorate)	HClO_4 (perchloric acid)
ClO^- (hypochlorite)	HClO (hypochlorous acid)
PO_4^{3-} (phosphate)	• H_3PO_4 (phosphoric acid)

- Note that SO_4^{2-} ion requires 2 H^+ ions to balance the charge, while the PO_4^{3-} ion requires 3 H^+ ions to balance its charge.

Binary molecular compounds (also known as binary covalent compounds) are substances composed of exactly two different nonmetals that are covalently bonded together.

How to Name Binary Molecular Compounds

- The first element is named the same as the element name.
- The second element we change its' ending to **-ide**.
- Greek prefixes** are used to indicate the number of atoms for each element. (The prefix *mono-* is not used for the first element.)
If the prefix ends in *a* or *o* and the name of the element starts with a vowel, the *a* or *o* of the prefix is often dropped.

These examples illustrate these rules:

Chemical Formula	Name	Chemical Formula	Name
N_2O	Dinitrogen monoxide	P_4S_{10}	Tetraphosphorous decasulfide
CO	Carbon monoxide	SO_3	Sulfur trioxide

Table 2.6 Prefixes Used in Naming Binary Molecular Compounds

Prefix	Meaning	Prefix	Meaning
<i>mono-</i>	1	<i>hexa-</i>	6
<i>di-</i>	2	<i>hepta-</i>	7
<i>tri-</i>	3	<i>octa-</i>	8
<i>tetra-</i>	4	<i>nona-</i>	9
<i>penta-</i>	5	<i>deca-</i>	10

A hydrate is a substance that contains water molecules as part of its chemical structure or crystal lattice. These water molecules are known as the “water of hydration”. The presence of water is indicated in the chemical formula by a centered dot ($\cdot$), followed by the number of water molecules per unit of the compound. For example, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ indicates five water molecules for every one unit of copper(II) sulfate. When heated, the water molecules can be driven off (removed by evaporation), leaving behind the substance in its anhydrous (without water) form. The anhydrous form often has different physical properties, such as a different color. For example, anhydrous cobalt(II) chloride is blue, while its hydrates form is magenta/red.

Hydrates are named by first naming the ionic compound, followed by a Greek prefix indication the number of water molecules, and then the word “hydrate”.

The following examples illustrate these rules:

Formula	Name
$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$	copper(II) sulfate pentahydrate
$\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$	Barium chloride dihydrate
$\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$	Iron(III) chloride hexahydrate

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3.2: Student Activity

Student Activity: In groups of 3-4, answer the following questions.

1. In each box, write the **chemical formula** of the compound formed by the cation at the head of the column and the anion at the left of the row, and then write the **name of the compound**.

I ons	Na^+	Hg^{2+}	NH_4^+	Pb^{2+}	Mg^{2+}	Fe^{3+}	Cu^{2+}
O x ide							
B r o m ide							
C h l o r ide							
C h l o r ide							
H y d r o x ide							
B r o m ide							
P h o s p h a t e							
I o d ide							
S u l f a t e							

2. In each box, write the chemical formula of the compound formed by the cation at the head of the column and the anion at the left of the row.

Ions	Potassium	Calcium	Chromium(III)	Zinc	Iron(III)	Aluminum
Nitrate						
Sulfate						
Hypochlorite						
Nitride						
Bromide						
Hydrogen phosphate						
Chloride						
Hydrogen carbonate						

3. In each box, when given a formula, write the name; given a name, write the formula.

Formula	Name	Name	Formula
SeO_4^{2-}		Perchloric acid	
$\text{HNO}_2(\text{aq})$		Cobalt(II) chloride hexahydrate	
$\text{Fe}(\text{NO}_3)_2$		Magnesium nitride	
$\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$		Selenic acid	
N_2O_5		Calcium sulfite	
PbO		Sodium hydride	
Cl_2O		Mercury(I) chloride	
$\text{Cr}(\text{ClO}_4)_3$		Nickel(II) sulfide	
$\text{K}_2\text{Cr}_2\text{O}_7$		Ammonium phosphate	
KMnO_4		Acetic acid	

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3.3: Assessment

Quiz – Naming and Chemical Formulas of Compounds

Instructions: You will be given 20 minutes to complete this quiz. Please answer each question to the best of your ability.

1. Fill in the following table as appropriate: [9 points]

Cation	Anion	Chemical Formula	Name of Compound
Cu^{2+}	PO_4^{3-}		
			Zinc Acetate
			Iron(III) Chromate
NH_4^+	CO_3^{2-}		
		AgMnO_4	
Ba^{2+}	ClO_2^-		
		SnO_2	

2. Fill in the following table as appropriate: [6 points]

Chemical Formula	Name of Compound
Si_2Br_6	
	Carbon tetrachloride
SO_3	
$\text{HClO}_4(aq)$	
	Sulfuric Acid
	Pentaphosphorus decoxide

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CHAPTER OVERVIEW

4: The Structure of the Atom

Introduction to Module

All matter is composed of atoms, the fundamental building blocks of chemical substances. Understanding the **structure of the atom** is essential for explaining how elements behave and interact in chemical reactions. An atom consists of a small, dense **nucleus** surrounded by electrons. The nucleus contains **protons**, which have a positive charge, and **neutrons**, which have no charge. **Electrons**, which carry a negative charge, move in regions around the nucleus called orbitals. The number of protons in an atom, known as the **atomic number**, identifies the element and determines its chemical properties.

Atoms of the same element can exist in different forms called **isotopes**. Isotopes have the same number of protons but different numbers of neutrons. Because of this difference, isotopes of an element have different **mass numbers** while maintaining the same chemical identity. Some isotopes are stable, while others are radioactive and decay over time.

The **atomic mass** of an element listed on the periodic table represents the **weighted average mass of all naturally occurring isotopes** of that element. Since different isotopes exist in different natural abundances, the atomic mass is typically not a whole number. Atomic mass is measured in **atomic mass units (amu)**, where one amu is defined relative to the mass of a carbon-12 atom.

The **periodic table** organizes all known elements according to increasing atomic number and recurring patterns in their chemical properties. Elements are arranged in rows called **periods** and columns called **groups**. Elements within the same group often share similar chemical behavior because they have similar electron arrangements. The periodic table allows chemists to predict trends in properties such as atomic size, reactivity, and bonding behavior.

Together, the concepts of atomic structure, isotopes, atomic mass, and the periodic table provide the foundation for understanding how elements are organized and how they interact to form the vast variety of substances found in nature.

Learning Outcomes

By the end of this module, students are expected to:

- Define atomic number and mass number and explain how they relate to the number of protons, neutrons, and electrons in an atom.
- Explain the concept of isotopes, describing how isotopes of the same element differ in mass number but have the same atomic number and similar chemical properties.
- Calculate the average atomic mass of an element using the masses and relative abundances of its isotopes.
- Interpret information from the periodic table, including atomic number, atomic mass, element symbols, and element classification.
- Identify trends and organization in the periodic table, including periods, groups, and the classification of elements as metals, nonmetals, and metalloids.
- Use periodic table information to predict basic properties of elements, such as relative atomic mass and common chemical behavior within groups.

[4.1: Atomic Number, Mass Number, and Atomic Symbol](#)

[4.2: Student Activity](#)

[4.3: Assessment](#)

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4.1: Atomic Number, Mass Number, and Atomic Symbol

Atomic Number, Mass Number, and Atomic Symbol

Every atom of a particular element has the same number of protons. This number is called the **atomic number** of the element. It is represented by the symbol **Z**. For example, all carbon atoms ($Z = 6$) have 6 protons, all nitrogen atoms ($Z = 7$) have 7 protons, and all lead atoms ($Z = 82$) have 82 protons. Atoms are electrically neutral; thus, the number of electrons is equal to the number of protons.

The **mass number**, represented by the symbol, **A**, is the total number of protons and neutrons in the nucleus of an atom. A carbon atom with 6 protons and 6 neutrons has a mass number of 12, and a chlorine atom with 17 protons and 18 neutrons has a mass number of 35.

The **atomic symbol** of an element is based on its English name for most elements. The symbol of an element is usually written using the first letter of the element, written using a capital letter. If more than one element begins with the same letter, a second letter, written in lowercase, is added. For example, the symbol for Carbon is C; however, the symbol for calcium is Ca. The symbol for hydrogen is H, whereas the symbol for helium is He. The symbols for most elements correspond to their English names, but the symbol of some elements are derived from their Latin names. For example, the symbol for sodium is Na, from its Latin name *natrium*, iron has the symbol Fe from *ferrum*, lead has the symbol Pb from *plumbum*.

The atomic symbol is usually written using the symbol of the element as well as the atomic number (Z) as a left *subscript* and the mass number (A) as a left *superscript*. So, in general, element X would be A_ZX .

Thus, a silicon atom, symbolized ${}^{14}_{14}\text{Si}$

Si, has $A = 29$, $Z = 14$, number of protons = 14, number of electrons = 14, and number of neutrons = $29 - 14 = 15$. Since each element has its own atomic number, we can figure out the atomic number given the element symbol. So, instead of writing ${}^{14}_{14}\text{Si}$ for silicon with mass number of 29, we can simply write

${}^{29}_{14}\text{Si}$ (silicon 29), with $Z = 14$ understood. Another way to name this atom is silicon-29.

Isotopes

Atoms of the same element that have different masses—different numbers of neutrons—are called **isotopes**. The term *isotope* refers to an atom of an element that has more than one variation in number of neutrons.

For example, there are 3 naturally occurring isotopes for carbon: C-12, C-13, and C-14. The most abundant is C-12, 98.89% of carbon has 6 neutrons.

The chemical properties of an element are primarily determined by the number of electrons, so *all isotopes of an element have nearly identical chemical behavior*, even though they have different masses.

There are 3 known isotopes for Hydrogen:

Isotope	Protons	Neutrons	Mass Number
Hydrogen-1 (Protium)	1	0	1
Hydrogen-2 (Deuterium)	1	1	2
Hydrogen-3 (Tritium)	1	2	3

Solved Example:

Argon has three naturally occurring isotopes, Ar-36, Ar-38, and Ar-40. What is the mass number of each? How many protons, neutrons, and electrons are present in each isotope?

Solution:

Using the Periodic table, we can find that the atomic number of Ar is 18.

Therefore:

Ar-36: mass number is 36; protons = 18, electrons = 18, neutrons $36 - 18 = 18$

Ar-38: mass number is 38; protons = 18, electrons = 18, neutrons $38 - 18 = 20$

Ar-40: mass number is 40; protons = 18, electrons = 18, neutrons $40 - 18 = 22$

The mass of a single atom is very small—much too small to be measured on a balance. The masses of atoms are typically not expressed in grams or kilograms. Instead, a very small mass unit has been defined that is more convenient for expressing the masses of atoms: the **atomic mass unit (amu)** is defined exactly as the mass of a carbon-12 atom: $1 \text{ amu} = 1.661 \times 10^{-24} \text{ g}$.

Samples of most pure elements consist of two or more isotopes. Thus, the mass of the atoms of an element is averaged based on the masses of the isotopes and their percentage abundance in nature. The **atomic mass** of an element is the average mass of all atoms of an element as they are distributed in nature. Keep in mind that atomic mass is always measured relative to the mass of an atom of carbon-12.

To find the **average atomic mass** of an element, you must know the atomic mass of each isotope and the fraction of each isotope in a sample.

We will use the following formula to calculate the average atomic mass:

$$\text{Average atomic mass} = \sum[(\text{isotope mass})(\text{fractional abundance})]$$

Solved Example:

Gallium has two naturally occurring, Ga-69 and Ga-71. Given the following data, calculate the average atomic mass of Ga.

Isotope	Mass (amu)	Abundance (%)
Ga-69	68.9256	60.11
Ga-71	70.9247	39.89

Solution:

First, you need to convert the % abundance to fractional abundances.

Fractional abundance of Ga-69 = $60.11/100 = 0.6011$; similarly, the fractional abundance of Ga-71 = $39.89/100 = 0.3989$.

Portion of atomic mass from Ga-69 = isotope mass $\times$ fractional abundance = $68.9256 \times 0.6011 = 41.43 \text{ amu}$

Portion of atomic mass from Ga-71 = isotope mass $\times$ fractional abundance = $70.9247 \times 0.3989 = 28.29 \text{ amu}$

Average atomic mass of Ga = $41.43 + 28.29 = 69.72 \text{ amu}$

Practice Problem:

Chlorine has two naturally occurring isotopes, Cl-35 (isotopic mass = 34.9689 amu), and Cl-37 (isotopic mass = 36.9659 amu). If chlorine has an average atomic mass of 35.4527 amu, what is the percent abundance of each isotope.

Solution:

Since chlorine has only two isotopes, the sum of the % abundances must equal 100, and this means that the sum of the fractional abundances is 1. Let's assume that the fractional abundance of Cl-35 is x , then the fractional abundance of Cl-37 is $1-x$.

$$\text{Average atomic mass of Cl} = \sum[(\text{isotope mass})(\text{fractional abundance})]$$

$$35.4527 = [(34.9689)(x)] + [(36.9659)(1-x)]$$

$$35.4527 = 34.9689x + 36.9659 - 36.9659x$$

Solve for x :

Subtract 36.9659 from both sides and add the two terms that contain x :

$$-1.5132 = -1.997x$$

Divide both sides by -1.997 :

$$x = 0.7577$$

So, the fractional abundance of Cl-35 = 0.7577; and the fractional abundance of Cl-37 = $1 - 0.7577 = 0.2423$. This means that Cl-35 = 75.55% and Cl-37 = 24.23%.

The Periodic Table is a tabular display of all known chemical elements, organized primarily by their atomic number (the number of protons in the nucleus). As of 2026, there are 118 confirmed elements, starting with Hydrogen (atomic number 1) and ending with Oganesson (atomic number 118).

The table is arranged into a grid of **periods** (horizontal rows) and **groups** (vertical columns). The table is made up of 7 rows and 18 columns. The groups are numbered 1 to 8 *and* either the letter A or B. A newer system, with group numbers from 1-18 but no letters can also be used. This was adopted by the International Union of Pure and Applied Chemistry (IUPAC). One common version of the modern periodic table appears in Figure 2.2 below.

The eight A groups (two on the left and six on the right) contain the **main-group elements**. The ten B groups, located between groups 2A(2) and 3A(13), contain the **transition elements**. Two horizontal series of **inner transition elements**, the *lanthanides* and *actinides*, fit between the elements in Group 3B(3) and group 4B(4) and are placed below the main body of the table.

Classifying the Elements

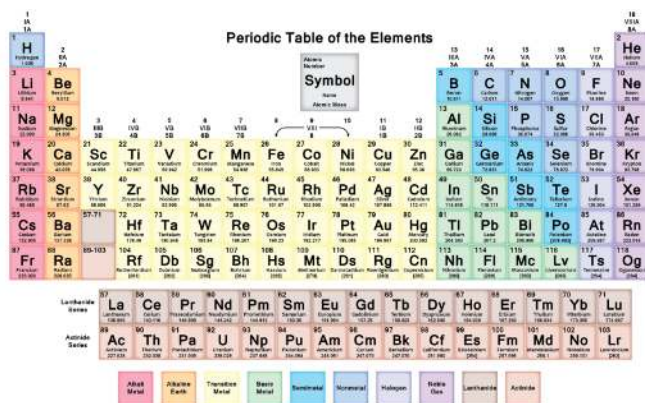
The elements can be classified as metals, nonmetals, and metalloids (or semimetals).

- **Metals:** lie in the left portion of the table. About three-quarters of the elements are metals, including many main-group elements and all the transition and inner transition elements. Metals are generally shiny solids at room temperature (mercury is the only liquid) that conduct heat and electricity well. They can be tooled into sheets (are malleable) and wires (are ductile).
- **Nonmetals:** lie in the small-upper portion of the table. They are generally gases or dull, brittle solids at room temperature (bromine is the only liquid) and conduct heat and electricity poorly.
- **Metalloids:** they lie between the metals and the nonmetals. They have properties between those of metals and nonmetals.

Common group names to know:

- Group 1A(1): alkali metals
- Group 2A(2): alkaline earth metals
- Group 6A(16): chalcogens
- Group 7A(17): halogens
- Group 8A(18): noble gases.

Periodic Table of the Elements



Legend:

- Alkali Metal
- Alkaline Earth
- Transition Metal
- Main Group Metal
- Metalloid
- Nonmetal
- Halogens
- Noble Gas
- Lanthanide
- Actinide

<https://sciencenotes.org/periodic-table-118-elements/>

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4.2: Student Activity

Student Activity

1. Using the information given below and your Periodic Table, fill in as many of the blanks as you can. If more information is needed, and you cannot fill the blank, leave the box empty or indicate using N/A. All atoms are neutral.

Name of Element	Atomic Symbol	Atomic Number	Mass Number	# of Protons	# of Neutrons	# of Electrons
					24	21
	Ge-76					
			122		72	
			37			17
Phosphorus			28			
		11			12	

2. A certain element consists of two stable isotopes. The first has an atomic mass of 68.9257 amu and a natural abundance percentage of 60.4%. The other isotope has an atomic mass of 70.9249 amu. Calculate the average atomic mass of the element. Using the Periodic Table, write its symbol and name.

3. Determine whether each statement that follows is true or false.

1. The atomic number of an element is the number of particles in the nucleus of an atom of that element.
2. An oxygen-16 atom has the same number of protons as an oxygen-17 atom.
3. The nuclei of nitrogen atoms have a different number of protons from the nuclei of any other element.
4. Isotopes of different elements that exhibit the same mass number exhibit similar chemical behavior.
5. Periods are arranged vertically in the periodic table.
6. Nb is the symbol of the element for which $Z = 41$.
7. Elements in the same column of the periodic table have similar properties.
8. The element for which $Z = 38$ is in both Group 2A/2 and the 5th period.
9. The difference among isotopes of an element is a difference in the number of neutrons in the nucleus.
10. All atoms of a specific element have the same number of protons.

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4.3: Assessment

For questions 1-6, circle the **CORRECT** answer: [1 point each]

- Which of these elements is most likely to be a good conductor of electricity?
A) N B) S C) He D) Cl E) Fe
- Which of the following is a metalloid?
A) carbon, C, $Z = 6$ B) sulfur, S, $Z = 16$ C) germanium, Ge, $Z = 32$
D) iridium, Ir, $Z = 77$ E) bromine, Br, $Z = 35$
- The elements in Group 2 are known by what name?
A) transition metals B) halogens C) alkali metals D) alkaline earth metals E) noble gases
- Which of these elements is chemically similar to oxygen?
A) sulfur B) calcium C) iron D) nickel E) potassium
- Which of the following is a nonmetal?
A) lithium, Li, $Z = 3$ B) bromine, Br, $Z = 35$ C) mercury, Hg, $Z = 80$
D) bismuth, Bi, $Z = 83$ E) sodium, Na, $Z = 11$
- In the periodic table, atoms are arranged in order of
A) increasing atomic mass. B) increasing atomic number. C) periodicity.
D) physical properties. E) chemical reactivities.

7. Fill in the blanks in the table: [5 points]

Symbol	Protons	Neutrons	Electrons	Mass Number
Hg-202				
	20	25		
			43	92

8. Determine the percent abundance of the two stable isotopes of silver, given the information below and the average atomic mass of silver is 107.9 amu. **Show your work.** [5 points]

Isotope	Mass of Isotope	Percent Abundance
Ag-107	106.9051 amu	?
Ag-109	108.9048 amu	?

9. The element rubidium has two naturally occurring isotopes. The atomic mass of Rb-85 (72.17 percent abundant) is 84.911794 amu. Determine the atomic mass of Rb-87. The average atomic mass of Rb is 85.4678 amu. [4 points]

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CHAPTER OVERVIEW

5: Chemical Reactions

Introduction to Module

Chemical reactions describe how substances change into new substances. Chemists represent these changes using **chemical equations**, which show the reactants (starting materials) and products (substances formed). Writing a correct chemical equation requires identifying the chemical formulas of the reactants and products and arranging them to represent the reaction that occurs.

Because matter is conserved in chemical reactions, chemical equations must follow the **law of conservation of mass**, which states that atoms are neither created nor destroyed during a reaction. To satisfy this law, equations must be **balanced**, meaning the number of atoms of each element is the same on both sides of the equation. Balancing chemical equations involves adjusting **coefficients** (numbers placed before formulas) so that the atoms of each element are conserved.

Once an equation is balanced, it can be used to perform **stoichiometric calculations**, which allow chemists to determine the quantitative relationships between reactants and products. The coefficients in a balanced equation represent the **mole ratios** that connect the amounts of substances involved in the reaction. These mole ratios make it possible to calculate how much product will form from a given amount of reactant or how much reactant is needed to produce a certain quantity of product.

In many reactions, the reactants are not present in the exact proportions required by the balanced equation. The **limiting reactant** is the reactant that is completely consumed first and therefore limits the amount of product that can be formed. Once the limiting reactant is used up, the reaction stops, even if other reactants remain in excess.

In practice, the amount of product obtained in an experiment is often less than the theoretical amount predicted by stoichiometric calculations. The **percent yield** compares the actual yield obtained in a reaction to the theoretical yield calculated from the balanced equation. It is expressed as a percentage and provides insight into the efficiency of a chemical reaction.

Together, writing and balancing chemical equations, performing stoichiometric calculations, identifying limiting reactants, and determining percent yield are essential skills in chemistry. These concepts allow chemists to predict reaction outcomes, measure reaction efficiency, and apply quantitative reasoning to chemical processes.

Learning Outcomes

By the end of this module, students are expected to:

- Define chemical reaction, chemical equation, reactant, and product.
- Write a chemical equation from a word reaction using chemical formulas for reactants and products.
- Balance chemical equations by changing stoichiometric coefficients.
- Determine amounts of reactant required or product formed using stoichiometry.
- Identify the limiting reactant in a reaction.
- Define theoretical yield, actual yield, and percent yield.
- Calculate the percent yield of a reaction.

[5.1: Stoichiometry of Formulas and Equations](#)

[5.2: Student Activity](#)

[5.3: Assessment](#)

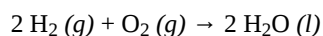
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5.1: Stoichiometry of Formulas and Equations

Writing and Balancing Chemical Equations

A **chemical reaction** is a process that neither creates nor destroys atoms, but it rearranges atoms in chemical compounds. A **chemical equation** uses chemical formulas to denote what occurs in a chemical reaction. A chemical equation is used to describe the chemical change that occurs when the original or starting substances, known as reactants, are converted to new substances, known as products. Reactants are written on the left side of the equation, whereas the products are on the right side. An arrow is used to indicate the direction of chemical change.

The chemical equation for the reaction of hydrogen gas with oxygen gas to form water is shown below:

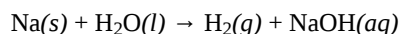


Atoms are neither created nor destroyed in a chemical reaction. To represent this in a chemical equation, coefficients may be used to balance the number of atoms of each element on both sides of the equation.

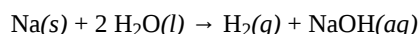
When writing a chemical equation, we must indicate the physical state of each substance. We use the symbol (s) for solids, (l) for liquids, (g) for gases, and (aq) for aqueous solutions—a homogeneous mixture in which a substance is dissolved in water.

According to the Law of Conservation of Mass, the total mass of the products of a reaction is the same as the total mass of the reactants. During a chemical reaction, atoms are neither created nor destroyed; they are simply rearranged to form new substances. In order to satisfy this law, equations must be balanced; an equation is balanced by placing **stoichiometric coefficients** in front of one or more of the formulas, indicating that it is used more than once.

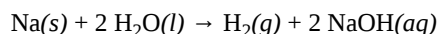
For example, if a piece of sodium metal is dropped into water, a vigorous reaction occurs, producing hydrogen gas and sodium hydroxide solution. That sentence can be translated literally into a chemical equation:



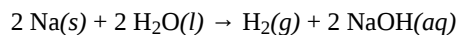
As we examine the above equation, we notice that there are only 2 H atoms on the left side, but 3 H atoms on the right. We can start by putting a coefficient of 2 in front of water:



Now, there are 4 H on the left and 3 on the right. Furthermore, O is now unbalanced (2 on the left and 1 on the right). Since O and H are both part of the same compound, NaOH, and there is one atom of each in that compound, if we add a coefficient of 2 in front of NaOH:



Now, both H and O are balanced: there are 4 H atoms on both sides of the equation, and 2 O atoms on both sides as well. However, Na is now not balanced, so we need to add a coefficient of 2 in front of Na:



The equation is now balanced. Note that in the absence of a numerical coefficient, as with H₂, the coefficient is assumed to be 1.

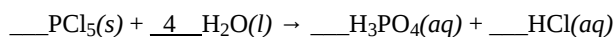
Solved Example:

Balance the equation for the reaction of solid phosphorus pentachloride with liquid water, yielding aqueous solutions of phosphoric acid and hydrochloric acid.

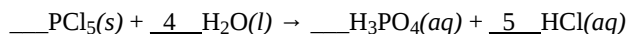
Translate the word reaction into a chemical equation using the chemical formula of each substance. Make sure to indicate the state of matter if given.



Start with the element(s) that are in only one other formula. For example, P and O are found only in one compound on the left, and one on the right. However, H is found in 2 compounds on the right, so we will balance H last. Since there is 1 P atom on both sides of the equation, P is already balanced. There is only 1 O atom on the left, but 4 on the right. So, we will add a coefficient of 4 in front of water:



Next, let's balance Cl. Since there are 5 Cl atoms on the left, but only 1 on the right, we will add a coefficient of 5 in front of HCl:



At this point, all elements are balanced: There are 1 P, 5 Cl, 8 H, and 4 O on each side of the equation.

Balancing a chemical equation requires something of a trail-and-error approach. In general, it will facilitate the balancing process if you do the following:

1. Change the coefficients of compounds (e.g., CO_2) before changing the coefficients of elements (e.g., O_2)
2. Treat polyatomic ions that appear on both sides of the equation as units, rather than counting their atoms individually.
3. Count atoms and/or polyatomic ions carefully, and track their numbers each time you change a coefficient.

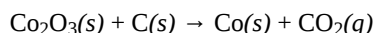
Balancing Chemical Equations Examples

EXAMPLE 3.1

Write a balanced equation for the reaction between solid cobalt(III) oxide and solid carbon to produce solid cobalt and carbon dioxide gas.

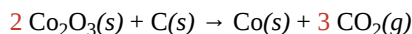
Solution:

1. Write a skeletal equation by writing chemical formulas for each of the reactant and products:



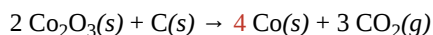
2. Balance atoms that occur in more complex substances first. Always balance atoms in compounds before atoms in pure substances:

We will begin with O: There are 3 O atoms on the left and 2 on the right. To balance O, put a 2 before Co_2O_3 and a 3 before CO_2

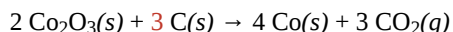


3. Balance atoms that occur as free elements on either side of the equation last. Always balance free elements by adjusting the coefficient on the free element.

Balance Co first: There are 4 Co on the left, and 1 Co on the right. To balance Co, put a 4 before $\text{Co}(s)$



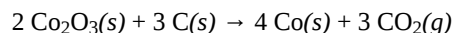
Balance C: There are 1 C on the left, and 3 on the right. To balance C, put a 3 before $\text{C}(s)$



4. If the balanced equation contains coefficient fractions, clear these by multiplying the entire equation by the denominator of the fraction.

This step is not necessary in this example.

5. Check to make sure the equation is balanced by summing the total number of each type of atom on both sides of the equation.



Left: 4 Co, 6 O, 3 C Right: 4 Co, 6 O, 3 C

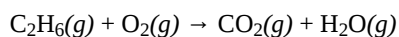
The equation is balanced.

EXAMPLE 3.2

Write a balanced equation for the combustion of gaseous ethane (C_2H_6), a minority component of natural gas, in which it combines with gaseous oxygen to form gaseous carbon dioxide and gaseous water.

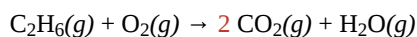
Solution:

1. Write a skeletal equation by writing chemical formulas for each of the reactant and products:

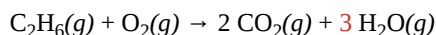


2. Balance atoms that occur in more complex substances first. Always balance atoms in compounds before atoms in pure substances:

We will begin with C: There are 2 C atoms on the left and 1 on the right. To balance C, put a 2 before $\text{CO}_2(g)$



We will then balance H: There are 6 H atoms on the left and 2 on the right. To balance H, put a 3 before $\text{H}_2\text{O}(g)$

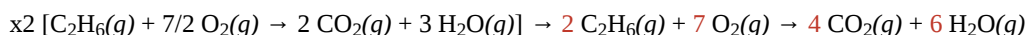


3. Balance atoms that occur as free elements on either side of the equation last. Always balance free elements by adjusting the coefficient on the free element.

We need to balance O: There are 2 O atoms on the left, and 7 on the right. To balance O, put a 7/2 before $\text{O}_2(g)$



4. If the balanced equation contains coefficient fractions, clear these by multiplying the entire equation by the denominator of the fraction.



5. Check to make sure the equation is balanced by summing the total number of each type of atom on both sides of the equation.



Left: 4 C, 12 H, 14 O Right: 4 C, 12 H, 14 O

The equation is balanced.

For Practice:

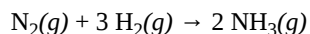
1. Write a balanced equation for the reaction between solid silicon dioxide and solid carbon to produce solid silicon carbide and carbon monoxide gas.
2. Write a balanced equation for the combustion of gaseous butane (C_4H_{10}), a fuel used in portable stoves and grills, in which it combines with gaseous oxygen to form gaseous carbon dioxide and gaseous water.
3. Solid sodium hydrogen carbonate reacts to form solid sodium carbonate, water, and carbon dioxide gas.

Calculating Quantities of Reactant and Product

Balanced chemical equations are very helpful tools to help us predict how much of a particular product will form from a given amount of a reactant, or measuring the amount of product formed, we can use this information to calculate the amount of reactant needed.

Moles of Reactants and Products

Based on the equation for the reaction of nitrogen gas with hydrogen gas to produce ammonia, NH_3 :



1 mole of N_2 combines with 3 moles of H_2 to produce 2 moles of NH_3 . We can write conversion factors using these **stoichiometric coefficients** to help us calculate the amount of NH_3 produced given any amount of N_2 or H_2 used, or how much of N_2 or H_2 is necessary to produce a specific amount of NH_3 .

The conversion factors can express the relationship between each reactant and the product, as well as the relationship between the two reactants. These can be written as

$$\frac{1 \text{ mol N}_2}{2 \text{ mol NH}_3} \quad \frac{3 \text{ mol H}_2}{2 \text{ mol NH}_3} \quad \frac{1 \text{ mol N}_2}{3 \text{ mol H}_2}$$

The ratio can also be written as the reciprocal:

$$\frac{2 \text{ mol } \text{NH}_3}{1 \text{ mol } \text{N}_2} \times \frac{2 \text{ mol } \text{NH}_3}{3 \text{ mol } \text{H}_2} \times \frac{3 \text{ mol } \text{H}_2}{1 \text{ mol } \text{N}_2}$$

Consider the complete reaction of 3.35 moles of N_2 to form NH_3 . To calculate the number of moles of NH_3 produced, we use the conversion factor with moles of NH_3 in the numerator and moles of N_2 in the denominator:

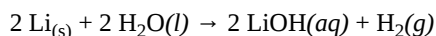
$$\text{moles } \text{NH}_3 \text{ produced} = 3.35 \text{ mol } \text{N}_2 \times \frac{2 \text{ mol } \text{NH}_3}{1 \text{ mol } \text{N}_2} = 6.70 \text{ mol } \text{NH}_3$$

We can determine the stoichiometric amount of H_2 (the exact number of moles of H_2 needed to react with 3.35 moles of N_2):

$$\text{moles } \text{H}_2 \text{ needed} = 3.35 \text{ mol } \text{N}_2 \times \frac{3 \text{ mol } \text{H}_2}{1 \text{ mol } \text{N}_2} = 10.1 \text{ mol } \text{H}_2$$

Example 3.3

How many moles of H_2 will be produced if 0.550 mol Li reacts according to the following equation?



Solution:

According to the balanced chemical equation, the conversion factor for H_2 and Li is either

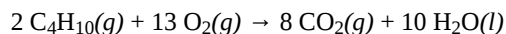
$$\frac{1 \text{ mol } \text{H}_2}{2 \text{ mol } \text{Li}} \text{ or } \frac{2 \text{ mol } \text{Li}}{1 \text{ mol } \text{H}_2}$$

To calculate the number of moles produced for H_2 , we choose the conversion factor in which H_2 is in the numerator:

$$\text{moles } \text{H}_2 \text{ produced} = 0.550 \text{ mol Li} \times \frac{1 \text{ mol } \text{H}_2}{2 \text{ mol Li}} = 0.275 \text{ mol } \text{H}_2$$

Practice:

1. Tetraphosphorus decoxide (P_4O_{10}) reacts with water to produce phosphoric acid. Write a balanced equation for this reaction and determine the number of moles of each reactant required to produce 7.50 moles of phosphoric acid.
2. Consider the combustion of butane (C_4H_{10}).



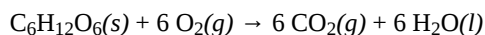
In a particular reaction, 5.0 moles of C_4H_{10} react with an excess of O_2 . Calculate the number of moles of CO_2 formed.

Mass of Reactants and Products

Balanced chemical equations give us the relative amounts of reactants and products in terms of moles. However, because we measure reactants and products in the laboratory by weighing them, most often such calculations start with mass rather than the number of moles.

Example 3.4

Calculate the mass of water produced by the metabolism of 65.5 g of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$), according to the following equation:



Solution:

We need to use the molar mass of $\text{C}_6\text{H}_{12}\text{O}_6$ to convert the given mass to moles, then use the appropriate stoichiometric conversion factor to convert moles of $\text{C}_6\text{H}_{12}\text{O}_6$ to moles of water, then using the molar mass of water, we convert to grams of water.

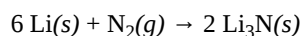
The molar mass are as follows: 180.16 g/mol for $\text{C}_6\text{H}_{12}\text{O}_6$ and 18.02 g/mol for H_2O . The conversion factor from $\text{C}_6\text{H}_{12}\text{O}_6$ to water is:

$$\frac{6 \text{ mol } \text{H}_2\text{O}}{1 \text{ mol } \text{C}_6\text{H}_{12}\text{O}_6}$$

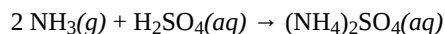
$$65.5 \text{ g } C_6H_{12}O_6 \times \frac{1 \text{ mol } C_6H_{12}O_6}{180.16 \text{ g } 1 \text{ mol } C_6H_{12}O_6} \times \frac{6 \text{ mol } H_2O}{1 \text{ mol } C_6H_{12}O_6} \times \frac{18.02 \text{ g } H_2O}{1 \text{ mol } H_2O} = 39.3 \text{ g } H_2O$$

Practice:

1. What mass of lithium nitride is produced when 252 g of lithium metal react with nitrogen according to the following equation?



2. The fertilizer ammonium sulfate $[(\text{NH}_4)_2\text{SO}_4]$ is prepared by the reaction between ammonia (NH_3) and sulfuric acid:



How many kilograms of NH_3 are needed to produce $1.00 \times 10^5 \text{ kg}$ of $(\text{NH}_4)_2\text{SO}_4$?

Limiting Reactant

In a chemical reaction, the **limiting reactant** (also called a limiting reagent) is **the substance that is completely consumed first**. Because it runs out before the other reactants, it determines the maximum amount of product that can be formed. Excess reactants are those present in quantities greater than necessary to react with the quantity of the limiting reactant.

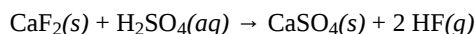
Determining the Limiting Reactant

In problems involving limiting reactants, the first step is to determine which reactant is the limiting reactant. In order to do this, we need to follow these steps:

1. Make sure the equation is balanced.
2. Calculate the number of moles of each reactant.
3. Using the stoichiometric coefficients, convert the number of moles of each reactant to moles (or grams) of product.
3. The reactant that produces the **least amount of product** is the limiting reactant.

Example 3.5

What mass of HF is produced according to the given equation when 5.00 g of each reactant are combined? What amount of excess reactant remains after the reaction is complete?



Solution:

Using its molar mass, convert 5.00 g of CaF_2 to moles of CaF_2 , then using the stoichiometric coefficients from the balanced equation, determine the number of moles of HF, and then using the molar mass of HF convert to mass of HF.

Follow the same steps above using the mass of H_2SO_4 . The reactant that produces the least amount of HF is the limiting reactant, and it will be the one used to determine how much of HF is produced.

General Procedure:

mass of reactant A $\rightarrow$ moles of A $\rightarrow$ moles of product $\rightarrow$ mass of product

$$5.00 \text{ g } \text{CaF}_2 \times \frac{1 \text{ mol } \text{CaF}_2}{78.08 \text{ g } \text{CaF}_2} \times \frac{2 \text{ mol HF}}{1 \text{ mol } \text{CaF}_2} \times \frac{20.01 \text{ g HF}}{1 \text{ mol HF}} = 2.56 \text{ g HF}$$

$$5.00 \text{ g } \text{H}_2\text{SO}_4 \times \frac{1 \text{ mol } \text{H}_2\text{SO}_4}{98.09 \text{ g } \text{H}_2\text{SO}_4} \times \frac{2 \text{ mol HF}}{1 \text{ mol } \text{H}_2\text{SO}_4} \times \frac{20.01 \text{ g HF}}{1 \text{ mol HF}} = 2.04 \text{ g HF}$$

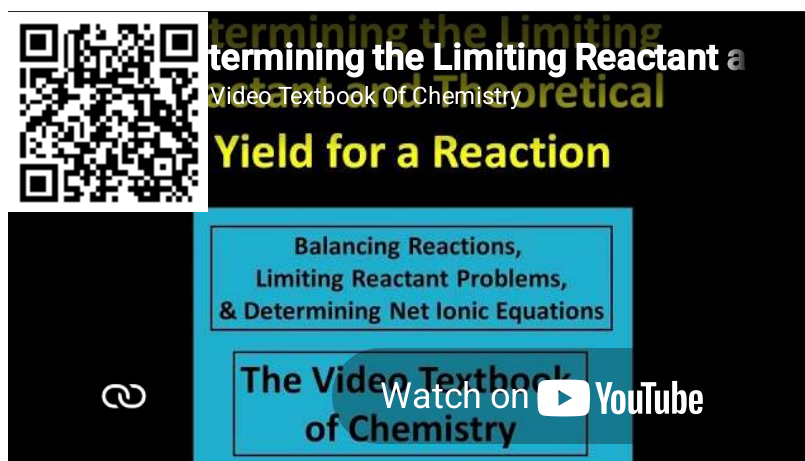
Since H_2SO_4 produced the **least amount of HF**, H_2SO_4 is the limiting reactant, and the amount of HF produced is 2.04 g.

Limiting reactant also determines how much of the other reactant is consumed in the chemical reaction. Using mass-mass conversion, use the mass of the limiting reactant given (H_2SO_4) to find the mass of CaF_2 used:

$$5.00 \text{ g } \text{H}_2\text{SO}_4 \times \frac{1 \text{ mol } \text{H}_2\text{SO}_4}{98.09 \text{ g } \text{H}_2\text{SO}_4} \times \frac{1 \text{ mol } \text{CaF}_2}{1 \text{ mol } \text{H}_2\text{SO}_4} \times \frac{78.08 \text{ g } \text{CaF}_2}{1 \text{ mol } \text{CaF}_2} = 3.98 \text{ g HF}$$

So the amount of HF used up in the reaction is 3.98 g. Excess amount of HF that remains after the reaction is complete is:

$$\text{excess amount HF} = 5.00 - 3.98 = 1.02 \text{ g HF}$$



Determining the Limiting Reactant and Theoretical Yield for a Reaction: <https://youtu.be/HmDm1qpNU0>

Practice:

1. Carbon dioxide can be removed from the air by reaction with potassium hydroxide: $\text{CO}_2 + 2 \text{KOH} \rightarrow \text{K}_2\text{CO}_3 + \text{H}_2\text{O}$. If 15.6 g of carbon dioxide reacts with 43.0 g of potassium hydroxide, what mass of potassium carbonate will be formed? What amount of excess reactant remains after the reaction is complete?
2. Aluminum nitrite and ammonium chloride react to form aluminum chloride, nitrogen gas, and water. How many grams of each substance are present after 72.5 g of aluminum nitrite and 58.6 g of ammonium chloride react completely.

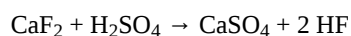
Theoretical, Actual, and Percent Yields

In example 3.5 above, we determined that the limiting reactant is H_2SO_4 and the amount of HF produced is 2.04 g. This is called the **theoretical yield**, the amount of product calculated based on the stoichiometry of the balanced chemical equation. In other words, the theoretical yield is the amount of product that forms when *all* the limiting reactant reacts to form the desired product. It is the maximum *obtainable* yield, predicted by the balanced equation. In practice, the **actual yield**-the amount of product actually obtained from a reaction-is almost always less than the theoretical yield. Chemists often determine the efficiency of a chemical reactions by calculating its percent yield, which tells *what percentage the actual yield is of the theoretical yield*. It is calculated as follows:

$$\% \text{ yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$$

Example 3.6

Hydrogen fluoride is used in the manufacture of Freons (which destroys ozone in the stratosphere) and in the production of aluminum metal. It is prepared by the reaction



In one process, 6.00 kg of CaF_2 are treated with an excess of H_2SO_4 to yield 2.86 kg of HF. Calculate the percent yield of HF.

Solution:

We are given that the actual yield is 2.86 kg of HF. Given the balanced chemical equation, and the initial mass of 6.00 kg of CaF_2 , you need to use the stoichiometric coefficients and determine the theoretical yield of HF.

Convert mass CaF_2 to moles CaF_2 using its molar mass, use the molar coefficients from the balanced equation to find moles of HF, and then the molar mass of HF convert to mass:

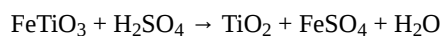
$$6.00 \text{ kg CaF}_2 \times \frac{1000 \text{ g}}{1 \text{ kg}} \times \frac{1 \text{ mol CaF}_2}{78.07 \text{ g CaF}_2} \times \frac{2 \text{ mol HF}}{1 \text{ mol CaF}_2} \times \frac{20.01 \text{ g HF}}{1 \text{ mol HF}} \times \frac{1 \text{ kg}}{1000 \text{ g}} = 3.08 \text{ kg HF}$$

$$\% \text{ yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$$

$$\% \text{ yield} = \frac{2.86 \text{ kg HF}}{3.08 \text{ kg HF}} \times 100\% = 92.9 \%$$

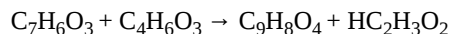
Practice:

1. Titanium(IV) oxide (TiO_2) is a white substance produced by the action of sulfuric acid on the mineral ilmenite (FeTiO_3):



Its opaque and nontoxic properties make it suitable as a pigment in plastics and paints. In one process, $8.00 \times 10^3 \text{ kg}$ of FeTiO_3 yielded $3.67 \times 10^3 \text{ kg}$ of TiO_2 . What is the percent yield of the reaction?

2. Aspirin, acetylsalicylic acid ($\text{C}_9\text{H}_8\text{O}_4$), is the most commonly used pain reliever in the world. It is produced by the reaction of salicylic acid ($\text{C}_7\text{H}_6\text{O}_3$) and acetic anhydride ($\text{C}_4\text{H}_6\text{O}_3$) according to the following equation:



In a certain aspirin synthesis, 104.8 g of salicylic acid and 110.9 g of acetic anhydride are combined. Calculate the percent yield of the reaction if 105.6 g of aspirin are produced.

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5.2: Student Activity

In groups of 3-4, answer the following questions. Show your work for full credit and use correct number of significant figures and units when necessary.

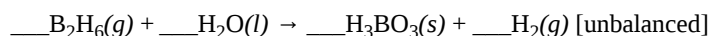
1. Convert the following into balanced equations:

- When lead(II) nitrate solution is added to potassium iodide solution, solid lead(II) iodide forms and potassium nitrate solution remains.
- Liquid disilicon hexachloride reacts with water to form solid silicon dioxide, hydrogen chloride gas, and hydrogen gas.
- When nitrogen dioxide is bubbled into water, a solution of nitric acid forms and gaseous nitrogen monoxide is released.

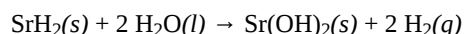
2. Write balanced equations for each of the following by inserting the correct coefficients in the blanks:

- $\text{P}_4\text{O}_{10}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{PO}_4(\text{l})$
- $\text{BCl}_3(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{BO}_3(\text{s}) + \text{HCl}(\text{g})$
- $(\text{CN})_2(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{C}_2\text{O}_4(\text{aq}) + \text{NH}_3(\text{g})$
- $\text{CaSiO}_3(\text{s}) + \text{HF}(\text{g}) \rightarrow \text{SiF}_4(\text{g}) + \text{CaF}_2(\text{s}) + \text{H}_2\text{O}(\text{l})$

3. Calculate the mass (g) of each product formed when 45.00 g of diborane, (B_2H_6), reacts with excess water, according to the following equation:



4. Metal hydrides react with water to form hydrogen gas and the metal hydroxide. For example:



You wish to calculate the mass (g) of hydrogen gas that can be prepared from 5.70 g of SrH_2 and 4.75 g of H_2O .

- What amount (mol) of H_2 can be produced from the given mass of SrH_2 ?
- What amount (mol) of H_2 can be produced from the given mass of H_2O ?
- Which is the limiting reactant?
- How many grams of H_2 can be produced?

5. a. Calculate the mass of H_2S that can form when 158 g of aluminum sulfide reacts with 131 g of water:



b. Calculate how many grams of the excess reactant remain after the reaction is complete.

6. When 20.5 g of methane and 45.0 g of chlorine gas undergo a reaction that has a 75.0% yield, what mass (g) of chloromethane (CH_3Cl) forms? Hydrogen chloride also forms.

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5.3: Assessment

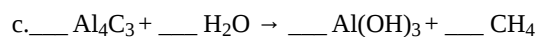
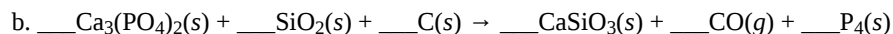
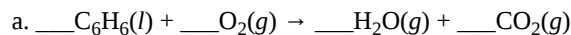
Instructions: You will be given **45 minutes** to complete this quiz. Please answer each question to the best of your ability using the correct number of significant figures and appropriate units when necessary. Proper work must be shown to receive full credit.

1. How many grams of calcium metal is required to react with 7.75 g of water to produce calcium hydroxide and hydrogen gas? Write and balance the chemical equation first. **[4 points]**

2. What mass of sodium fluoride (used in water fluoridation and the manufacture of insecticides) is required to form 485 g of sulfur tetrafluoride? **[4 points]**



3. Balance the following equations: **[7 points]**

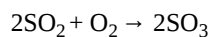


4. The first step in the Ostwald process for producing nitric acid is as follows: **[5 points]**



If the reaction of 15.0 g of ammonia with 15.0 g of oxygen gas yields 8.70 g of nitric oxide, what is the percent yield of this reaction?

5. What mass of excess reactant remains at the end of the reaction if 90.0 g of SO_2 are mixed with 100.0 g of O_2 ? **[5 points]**



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CHAPTER OVERVIEW

6: Three Major Classes of Chemical Reactions

Introduction to Module

Chemical reactions in aqueous solutions are commonly classified into several important types based on how substances behave when dissolved in water and how atoms or ions rearrange during reactions. Understanding these reaction types helps chemists predict products, write balanced equations, and analyze chemical processes in laboratory and real-world systems.

When substances dissolve in water, they may or may not produce ions. **Electrolytes** are substances that dissolve in water to produce ions, allowing the solution to conduct electricity. Electrolytes are typically soluble ionic compounds (such as salts) and certain molecular compounds like strong acids and bases. Strong electrolytes dissociate completely into ions, while weak electrolytes only partially dissociate. In contrast, **nonelectrolytes** dissolve in water without forming ions; instead, they remain as intact molecules. Because they do not produce ions, nonelectrolyte solutions do not conduct electricity. Examples include many sugars and alcohols.

A **precipitation reaction** occurs when two aqueous ionic solutions react to form an insoluble solid called a **precipitate**. During these reactions, dissolved ions exchange partners (a double-replacement reaction). If one of the products is insoluble in water, it separates from the solution as a solid. Solubility rules are commonly used to predict whether a precipitate will form. Writing **complete ionic equations** and **net ionic equations** helps identify the actual chemical change by removing spectator ions—ions that remain unchanged during the reaction.

Acid–base reactions involve the transfer of protons (H^+ ions) between substances. According to the Brønsted–Lowry definition, **acids donate protons** while **bases accept protons**. In aqueous solutions, many acid–base reactions are **neutralization reactions**, where an acid reacts with a base to produce water and a salt. Strong acids and bases dissociate completely in water, while weak acids and bases only partially ionize. Acid–base reactions play an essential role in many chemical, biological, and environmental processes.

Oxidation–reduction reactions, or **redox reactions**, involve the transfer of electrons between species. **Oxidation** refers to the loss of electrons, while **reduction** refers to the gain of electrons. These processes always occur together: when one substance is oxidized, another is reduced. Oxidation states are often used to track electron transfer and identify which species undergo oxidation or reduction. Redox reactions are fundamental to many important processes, including combustion, corrosion, metabolism, and electrochemical reactions such as batteries.

Learning Outcomes

By the end of this module, students are expected to:

- Categorize compounds as nonelectrolytes, weak electrolytes, or strong electrolytes.
- Identify weak and strong acids and bases.
- Apply solubility guidelines toward determining whether a reaction will produce a precipitate.
- Produce the molecular, ionic, and net ionic equations for a reaction.
- Identify the spectator ions in a reaction.
- Understand the definition of Arrhenius and Brønsted acids and bases.
- Predict the neutralization reaction between an acid and a base.
- Identify the various components of an oxidation-reduction reaction including reducing/oxidizing agents and half-reactions.
- Apply oxidation number rules toward determining the oxidation number of each element in a compound or polyatomic ion.
- Utilize the activity series to determine whether a reaction occurs between a metal and ions of another metal.
- Identify the different types of reactions that may be oxidation-reduction reactions, including combination, decomposition, and combustion.

[6.1: Three Major Classes of Chemical Reactions](#)

[6.2: Student Activity](#)

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6.1: Three Major Classes of Chemical Reactions

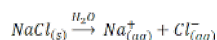
General Properties of Aqueous Solutions

Chemical reactions involving reactions dissolved in water are among the most common and important. A **solution** is a homogeneous mixture of two or more substances. Solutions may be gaseous (such as air), solid (such as brass), or liquid (such as salt-water). The majority component of the mixture is the **solvent**, and the substance present in smaller amount is the **solute**. An **aqueous solution** is one in which the solvent is water.

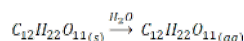
Electrolytes and Nonelectrolytes

You have probably heard of electrolytes in the context of sports drinks such as Gatorade. Electrolytes are necessary for the transmission of electrical impulses, which are critical to physiological processes such as nerve impulses and muscle contractions. In general, an **electrolyte** is a substance that dissolves in water to yield a solution that conducts electricity. On the other hand, a **nonelectrolyte** is a substance that dissolves in water but does not form ions. Because no ions are produced, the solution that *does not* conduct electricity.

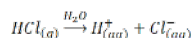
When an ionic compound dissolves in water, it *dissociates*, producing separate aqueous ions. The presence of these ions in solution is what makes the solution conduct electricity. For example, when sodium chloride (NaCl) dissolves in water, it dissociates to produce the sodium ions and chloride ions. **Dissociation** is the process by which an *ionic* compound, upon dissolution, breaks apart into its *constituents* ions. **Ionic compounds** that are water-soluble, are **electrolytes**.



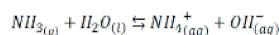
However, when a compound such as sucrose dissolves in water, the sucrose molecules remain intact upon dissolving, becoming aqueous sucrose *molecules*.



Molecular compounds that are water-soluble, such as sucrose, are **nonelectrolytes**. The molecular compounds that are electrolytes include *acids* and *molecular bases*. Acids and molecular bases undergo ionization in solution. **Acid** is a substance that produces *hydrogen ions* (H^+) when dissolved in water. **Ionization** is the process by which a molecular compound forms aqueous ions when it dissolves. HCl, for example, ionizes to produce aqueous H^+ ions and Cl^- ions.



A **base** is a compound that produces *hydroxide ions* (OH^-) when dissolved in water. Ammonia (NH_3), for example, ionizes in water to produce hydroxide (OH^-) and ammonium (NH_4^+) ions.



Strong Electrolytes and Weak Electrolytes

A **strong electrolyte** is a substance when dissolves in water it *dissociates completely*, which means all of the dissolved compound exists in the form of ions. **All water-soluble ionic compounds are strong electrolytes** since they dissociate completely upon dissolving. Figure 7.1 shows a comparison between a strong electrolyte, a weak electrolyte, and a nonelectrolyte.

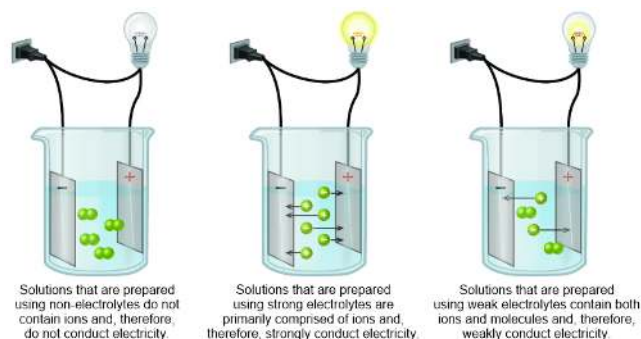


Figure 6.1: The electrical conductivity of solutions that contain non-electrolytes, strong electrolytes, and weak electrolytes, respectively. (Image: by Anonymous via LibreTexts, CC BY-NC-SA 3.0)

<https://chem.libretexts.org/Courses/...ce=chatgpt>

The only **molecular compounds that are strong electrolytes are the 7 strong acids**, which are listed in Table 6.1. A strong acid ionizes completely, resulting in a solution that contains hydrogen ions and the corresponding anions but essentially no acid molecules.

Acid	Ionization Equation
Hydrochloric acid	$\text{HCl}_{(aq)} \rightarrow \text{H}^+_{(aq)} + \text{Cl}^-_{(aq)}$
Hydrobromic acid	$\text{HBr}_{(aq)} \rightarrow \text{H}^+_{(aq)} + \text{Br}^-_{(aq)}$
Hydroiodic Acid	$\text{HI}_{(aq)} \rightarrow \text{H}^+_{(aq)} + \text{I}^-_{(aq)}$
Nitric Acid	$\text{HNO}_{3(aq)} \rightarrow \text{H}^+_{(aq)} + \text{NO}_3^-_{(aq)}$
Chloric Acid	$\text{HClO}_{3(aq)} \rightarrow \text{H}^+_{(aq)} + \text{ClO}_3^-_{(aq)}$
Perchloric acid	$\text{HClO}_{4(aq)} \rightarrow \text{H}^+_{(aq)} + \text{ClO}_4^-_{(aq)}$
Sulfuric Acid*	$\text{H}_2\text{SO}_{4(aq)} \rightarrow \text{H}^+_{(aq)} + \text{HSO}_4^-_{(aq)}$
	$\text{HSO}_4^-_{(aq)} \rightleftharpoons \text{H}^+_{(aq)} + \text{SO}_4^{2-}_{(aq)}$

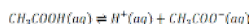
Strong bases are the alkali metal hydroxides, including CsOH, NaOH, KOH, RbOH, and CsOH, and the heavy alkaline earth metal hydroxides, including $\text{Ca}(\text{OH})_2$, $\text{Sr}(\text{OH})_2$, and $\text{Ba}(\text{OH})_2$.

A **weak electrolyte** is a substance that **partially ionizes (dissociates) in water**, producing only a **small number of ions** in solution. Because there are fewer ions available to carry electric charge, weak electrolytes **conduct electricity poorly** compared to strong electrolytes.



Figure 6.2: The Difference between Soluble and Insoluble Compounds (a) and Strong, Weak, and Nonelectrolytes (b). When a soluble compound dissolves, its constituent atoms, molecules, or ions disperse throughout the solvent. In contrast, the constituents of an insoluble compound remain associated with one another in the solid. A soluble compound is a strong electrolyte if it dissociates completely into ions, a weak electrolyte if it dissociates only slightly into ions, and a nonelectrolyte if it dissolves to produce only neutral molecules. (Image: by Anonymous via LibreTexts, CC BY-NC-SA 3.0) https://chem.libretexts.org/Bookshelves/Aqueous_Solutions

Weak acids and weak bases are examples of weak electrolyte. For example, acetic acid (CH_3COOH) is an example of a weak acid. Acetic acid only ionizes partially, resulting in only a small percentage of ions present in solution:



The double-sided arrow indicates a **reversible reaction**. It means the reaction can proceed in **both directions**: reactants form products, and the products can also react to reform the reactants. Eventually the system may reach **chemical equilibrium**, where both reactions occur at the same rate.

Precipitation Reactions

A **precipitation reaction** occurs when two soluble ionic compounds are mixed, and one of the products is insoluble in water. A precipitation reaction is the formation of an insoluble product through the net removal of ions from solution. This insoluble solid product is called a **precipitate**. Precipitation usually involve ionic compounds, but only certain combinations of electrolyte solutions result in the formation of a precipitate. To predict whether a precipitate will form when we mix two aqueous ionic solutions, we will refer to the solubility guidelines in Tables 6.2 and 6.3.

Solubility Guidelines

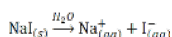
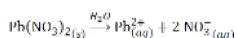
Table 6.2 Solubility Guidelines: Soluble Compounds

Water-Soluble Compounds	Insoluble Exceptions
Compounds containing an alkali metal cation (Li^+ , Na^+ , K^+ , Rb^+ , Cs^+) or the ammonium ion (NH_4^+)	
Compounds containing the nitrate ion (NO_3^-), acetate ion ($\text{C}_2\text{H}_3\text{O}_2^-$), or chlorate ion (ClO_3^-)	
Compounds containing the chloride ion (Cl^-), bromide ion (Br^-), or iodide ion (I^-)	Compounds containing Ag^+ , Hg_2^{2+} , or Pb^{2+}
Compounds containing the sulfate ion (SO_4^{2-})	Compounds containing Ag^+ , Hg_2^{2+} , Pb^{2+} , Ca^{2+} , Sr^{2+} , or Ba^{2+}

Table 6.3 Solubility Guidelines: Insoluble Compounds

Water-Insoluble Compounds	Soluble Exceptions
Compounds containing the carbonate (CO_3^{2-}), phosphate ion (PO_4^{3-}), chromate ion (CrO_4^{2-}), or sulfide ion (S^{2-})	Compounds containing alkali metal (Li^+ , Na^+ , K^+ , Rb^+ , Cs^+) or NH_4^+
Compounds containing the hydroxide ion (OH^-)	Compounds containing Li^+ , Na^+ , K^+ , Rb^+ , Cs^+ , Ca^{2+} , Sr^{2+} , or Ba^{2+}

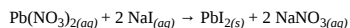
When lead(II) nitrate and sodium iodide are each dissolved in water, they form soluble ionic compounds, based on the solubility guidelines. This makes them strong electrolytes which means they will dissociate completely in solution. We can write the dissociation equations for each as follows:



In order to predict if a precipitate forms, first we have to note all the ions present in the solutions. We then must consider the cation-anion combinations that will result with an insoluble compound. In addition to $\text{Pb}(\text{NO}_3)_2$ and NaI , which we know are soluble, the other cation-anion combinations are PbI_2 and NaNO_3 . According to Table 6.2, PbI_2 is not soluble in water, and thus it will form a **precipitate**.

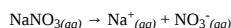
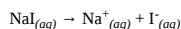
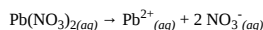
Molecular, Total Ionic, and Net-Ionic Equations

When two ionic compounds are mixed, the ions exchange partners, i.e., the metal cations seems to exchange anions. Based on the above chemical equation, the Pb^{2+} ion originally with NO_3^- ions, ends up paired with I^- ions; similarly, each Na^+ ion, originally paired with an I^- ion, ends up paired with an NO_3^- ion. Such reactions are called **double-displacement reactions**, or **metathesis reactions**.

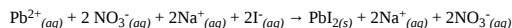


This equation, as written, is called a **molecular equation**, which is a chemical equation where all reactants and products are represented by their chemical formulas, in their undissociated form, making them look as if they exist in solution as molecules or formula units.

As mentioned above, soluble ionic compounds are **strong electrolytes**. They exist in solution as hydrated **ions**, rather than formula units. Thus it would be more realistic to represent the aqueous species in the reaction as follows:

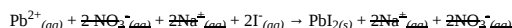


So rewriting the equation representing the ionic compounds as hydrated ions, it would be as follows:

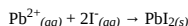


This is called the **total ionic equation**, which is a more accurate because it shows all the soluble ionic compounds (strong electrolytes) dissociated as ions. Insoluble compounds, such as PbI_2 , will be represented with their chemical formulas.

Notice that $\text{Na}^+(\text{aq})$ and $\text{NO}_3^-(\text{aq})$ appear unchanged on both sides of the equation. These are called **spectator ions**. These ions are not involved in the actual chemical change and thus do not participate in the reaction. Spectator ions cancel one another, just as identical terms on both sides of an algebraic equation cancel one another.



Eliminating the spectator ions yields the following equation:



This is called the **net ionic equation**, a chemical equation that includes only the species that are actually involved in the chemical change. The net ionic equation tells us what actually happens when we combine solutions together.

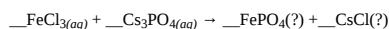
Sample example 6.1

Predict whether a reaction occurs, and if so, write balanced molecular, total ionic, and net ionic equations for each of the following combinations:

- (a) iron(III) chloride(aq) + cesium phosphate(aq) →
 (b) magnesium bromide(aq) + potassium acetate(aq) →

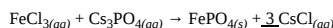
Solution:

(a) We predict the products by exchanging ions. Fe^{3+} ion will exchange the Cl^{-} ion for PO_4^{3-} ; and similarly, the Cs^{+} ion will exchange the PO_4^{3-} ion for the Cl^{-} . We must write a balanced chemical equation and based on the solubility guidelines, predict whether an insoluble product forms.



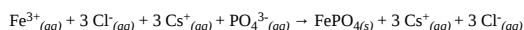
Based on Table 6.2, compounds containing an alkali metals form soluble compounds with no exception, that's why we will predict that CsCl is a soluble ionic compound. However, based on Table 6.3, compounds containing phosphates are usually insoluble except when combined with an alkali metal or ammonium, which is not this case, and so we will predict that FePO_4 will form as a precipitate.

Balanced Molecular Equation:

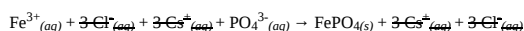


Total Ionic Equation:

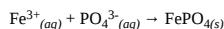
All soluble ionic compounds are strong electrolytes, and will need to write them in their ionic form, solids do not form ions:



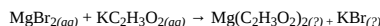
We need to identify the spectator ions and cancel them from both sides of the equation. Note that Cl^{-} and Cs^{+} ions exist in both sides of the equation in the same form, this indicates that they did not go through any chemical change, and these are the spectator ions.



Net Ionic Equation:

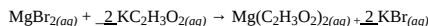


- (b) When magnesium bromide reacts with potassium acetate, the Mg^{2+} ion exchange the Br^{-} ion for the $\text{C}_2\text{H}_3\text{O}_2^{-}$ and K^{+} ion will react with the Br^{-} ion according to the following equation:



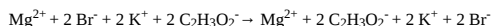
Based on the solubility guidelines, *compounds containing acetates and alkali metals always form soluble compounds*, and so both products in this reaction are soluble.

Balanced molecular equation:



Total ionic equation:

All are soluble ionic compounds, so they will exist in solution as ions:



Net ionic equation:

Since all compounds are soluble, all of these ions are spectator ions, they will cancel out, so **there is no reaction**.

Acid-Base Reactions (Neutralization Reaction)

Acid-base reactions are among the most fundamental processes in chemistry and occur in **biological systems, industrial chemistry, environmental chemistry, and analytical methods**. These reactions involve the **transfer of protons (H^{+})** or the **interaction of species that donate or accept electron pairs**, depending on the theoretical framework used.

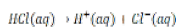
Arrhenius Definition (Most Basic)

Arrhenius Acid: produces H^{+} ions in water

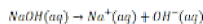
Arrhenius Base: produces OH^{-} ions in water

Examples:

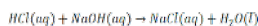
Acid:



Base:



Neutralization example:

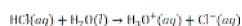


Brønsted-Lowry Definition (More General)

Brønsted-Lowry Acid: proton donor

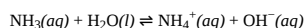
Brønsted-Lowry Base: proton acceptor

Example 1 (Water as a base):



- HCl donates a proton → acid
- H_2O accepts a proton → base

Example 2 (Water as an acid):



- **H₂O** donates a proton → acid
- **NH₃** accepts a proton → base

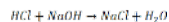
Water can act as **either an acid or a base**. Such substances are called **amphoteric**.

Acid–Base Reactions (Neutralization Reactions)

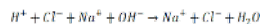
A **neutralization reaction** occurs when an acid reacts with a base to produce **water and a salt**.

Example:

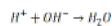
Molecular Equation



Complete Ionic Equation



Net Ionic Equation



Spectator ions (Na⁺ and Cl[−]) cancel.

This reaction forms the basis of **acid–base titrations**.

Oxidation-Reduction (Redox) Reactions

Oxidation–reduction (redox) reactions are chemical reactions involving the **transfer of electrons between species**. These reactions are central to many chemical and biological processes, including cellular respiration, photosynthesis, corrosion of metals, batteries and electrochemistry, and industrial metallurgy.

In any redox reaction, **one substance loses electrons while another substance gains electrons**. Because electrons must be conserved, oxidation and reduction always occur **simultaneously**.

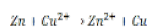
Oxidation is defined as the loss of electrons. **Reduction** is the gain of electrons

A helpful mnemonic is:

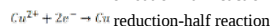
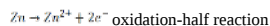
OIL RIG

- **Oxidation Is Loss**
- **Reduction Is Gain**

Example:



Electron transfer:



- Zinc **loses electrons** → Zn is oxidized
- Copper ion **gains electrons** → Cu²⁺ is reduced

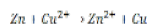
Oxidizing and Reducing Agents

In a redox reaction, two important roles exist.

Reducing Agent: Substance that **donates electrons** and becomes oxidized.

Oxidizing Agent: Substance that **accepts electrons** and becomes reduced.

Example:



Zn: reducing agent; Cu²⁺ is the oxidizing agent.

Oxidation Numbers (Oxidation States)

Oxidation numbers are a **bookkeeping system used to track electron transfer**.

They represent the **hypothetical charge an atom would have if electrons were transferred completely**. Table 6.5 shows the rules for assigning oxidation numbers.

Table 6.5 Rules for Assigning an Oxidation Number

General Rules

1. The oxidation number of an atom in its elemental form is zero.	Na, O ₂ , He, N ₂ , Fe
2. The oxidation number of a monatomic ion is equal to the charge on the ion.	For example, in the compound FeSO ₄ , the O.N. for Fe is +2 since the ions present are Fe ²⁺ and SO ₄ ²⁻
3. The sum of O.N. for all atoms in a molecule is zero.	For example, for the H ₂ O molecule, the sum of O.N. of all atoms = zero
4. The sum of the O.N. for all atoms in a polyatomic ion is equal to the charge on the ion.	For example, for the SO ₄ ²⁻ ion, the sum of the O.N. for all atoms is equal to -2. The sum of O.N. for all atoms in the PO ₄ ³⁻ ion is equal to -3.

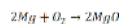
Rules for Specific Elements or Periodic Table Group

Element/Group	Oxidation Number	Exceptions
Fluorine	Always -1 in all of its compounds	
Group 1A or 2A	Group 1A elements always have +1 O.N. Group 2A elements always have +2 O.N.	
Hydrogen	Always +1 when combined with nonmetals	When combined with elements in Group 1A or 2A, hydrogen will have O.N. of -1 Example: the O.N. for H in NaH is -1
Oxygen	Most common O.N. for oxygen is -2	When Oxygen is combined with anything higher on this list (with F, group 1A, 2A elements, or H), you must calculate the O.N. for oxygen following the above general rules (see rule 3 above). Examples: the O.N. for O in OF ₂ is +2; the O.N. for O in H ₂ O ₂ is -1; the O.N. for O in KO ₂ is -1/2
Group 7A (other than fluorine)	O.N. for halogens is usually -1, especially when combined with metals	Any combination with something higher on the list will necessitate the halogen a different O.N. (see rules 3 and 4 above). Examples, the O.N. for Cl in ClF is +1; the O.N. for Br in BrO ₃ ⁻ is +5; the O.N. for I in IO ₄ ⁻ is +7

Note: The proper convention for reporting charge is to write the number first, followed by the sign (e.g., 2+), while oxidation number is written with the reversed sequence, sign followed by number (e.g., +2). This convention aims to emphasize the distinction between these two related properties.

Identifying Redox Reactions Using Oxidation Numbers (O.N.)

Example:



Oxidation numbers:

Species	Oxidation Number
Mg	0
O ₂	0
Mg in MgO	+2
O in MgO	-2

Changes:

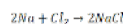
- Mg: 0 → +2 (**oxidation**)
- O: 0 → -2 (**reduction**)

Types of Redox Reactions

1. Combination (Synthesis)

Two substances combine to form one product.

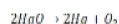
Example:



2. Decomposition

One compound breaks into simpler substances.

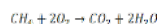
Example:



3. Combustion Reactions

Reaction with oxygen producing oxides.

Example:

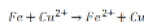


Carbon O.N. changes from -4 to +4, C is oxidized; oxygen O.N. changes from 0 to -2, O is reduced.

4. Single Displacement Reactions

One element replaces another.

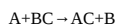
Example:



Common with **metal activity series**.

A **single-displacement reaction** (also called **single-replacement reaction**) occurs when **one element replaces another element in a compound**. These reactions are typically **redox reactions**, because electrons are transferred between the reacting species.

General form:



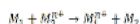
Where:

- **A** = free element
- **BC** = compound
- **AC** = new compound
- **B** = displaced element

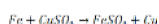
For the reaction to occur, the **free element must be more reactive than the element it replaces**.

1. Metal Displacing Another Metal

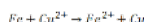
General form:



Example:



Ionic form:



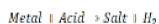
Explanation:

- Iron is **more reactive** than copper
- Iron **loses electrons**
- Copper ions **gain electrons**

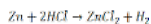
2. Metal Displacing Hydrogen from Acids

Some metals react with acids to produce **hydrogen gas**.

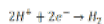
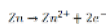
General reaction:



Example:



Half-reactions:



Observation:

- Bubbling **hydrogen gas**
- Metal dissolves

The **activity series** (Table 6.6) ranks metals according to their **tendency to lose electrons (be oxidized)**.

Metals **higher in the series are more reactive** and can displace metals below them.

Table 6.6 The Activity Series

	Element	Oxidation Reaction
React vigorously with cold H ₂ O to form H ₂	Lithium	$\text{Li} \rightarrow \text{Li}^+ + \text{e}^-$
	Potassium	$\text{K} \rightarrow \text{K}^+ + \text{e}^-$
	Barium	$\text{Ba} \rightarrow \text{Ba}^{2+} + 2\text{e}^-$
	Calcium	$\text{Ca} \rightarrow \text{Ca}^{2+} + 2\text{e}^-$
	Sodium	$\text{Na} \rightarrow \text{Na}^+ + \text{e}^-$
React with steam to form H ₂	Magnesium	$\text{Mg} \rightarrow \text{Mg}^{2+} + 2\text{e}^-$
	Aluminum	$\text{Al} \rightarrow \text{Al}^{3+} + 3\text{e}^-$
	Manganese	$\text{Mn} \rightarrow \text{Mn}^{2+} + 2\text{e}^-$
	Zinc	$\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$
	Chromium	$\text{Cr} \rightarrow \text{Cr}^{3+} + 3\text{e}^-$
React with simple acids to form H ₂	Iron	$\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$
	Cadmium	$\text{Cd} \rightarrow \text{Cd}^{2+} + 2\text{e}^-$
	Cobalt	$\text{Co} \rightarrow \text{Co}^{2+} + 2\text{e}^-$
	Nickel	$\text{Ni} \rightarrow \text{Ni}^{2+} + 2\text{e}^-$
	Tin	$\text{Sn} \rightarrow \text{Sn}^{2+} + 2\text{e}^-$
Will not dissolve in simple acids	Lead	$\text{Pb} \rightarrow \text{Pb}^{2+} + 2\text{e}^-$
	Hydrogen	$\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$
	Copper	$\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$
	Silver	$\text{Ag} \rightarrow \text{Ag}^+ + \text{e}^-$
	Mercury	$\text{Hg} \rightarrow \text{Hg}^{2+} + 2\text{e}^-$
	Platinum	$\text{Pt} \rightarrow \text{Pt}^{2+} + 2\text{e}^-$
	Gold	$\text{Au} \rightarrow \text{Au}^+ + \text{e}^-$

Increasing ease of oxidation

https://chem.libretexts.org/Bookshelves/General_Chemistry/Bookshelves/General_Chemistry/10%3A%20Acid-Base%20Equilibria/10.1%3A%20The%20Activity%20Series

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6.2: Student Activity

In groups of 3-4, answer the following questions.

1. Refer to the Solubility Rules in your textbook and indicate whether the following compounds are soluble or insoluble in water:

- (a) NH_4Cl _____ (b) $\text{Fe}(\text{C}_2\text{H}_3\text{O}_2)_3$ _____ (c) $\text{Mg}(\text{NO}_3)_2$ _____
 (d) AgI _____ (e) BaSO_4 _____ (f) CaCO_3 _____
 (g) CuCrO_4 _____ (h) AlPO_4 _____ (i) ZnS _____
 (j) $\text{Sr}(\text{OH})_2$ _____ (k) Na_3PO_4 _____ (l) CaS _____

2. For each of the following pairs of aqueous solutions, state whether a precipitation reaction occurs when they are mixed. Write the names and formulas of any precipitates that form.

Reactants	Will a precipitate form?	Name & Formula of Precipitate
a) Ammonium chloride + Silver Nitrate		
b) Sodium nitrate + nickel (II) sulfate		
c) cesium carbonate + Barium Hydroxide		

3. Classify the following types of chemical reactions:

- a) $2\text{Na}_{(s)} + \text{Cl}_{2(g)} \rightarrow 2\text{NaCl}_{(s)}$
 b) $\text{Na}_2\text{CO}_{3(s)} \xrightarrow{\Delta} \text{Na}_2\text{O}_{(s)} + \text{CO}_{2(g)}$
 c) $2\text{Na}_{(s)} + 2\text{H}_2\text{O}_{(l)} \rightarrow 2\text{NaOH}_{(aq)} + \text{H}_{2(g)}$
 d) $\text{AgNO}_{3(aq)} + \text{NaCl}_{(aq)} \rightarrow \text{AgCl}_{(s)} + \text{NaNO}_{3(aq)}$
 e) $\text{HNO}_{3(aq)} + \text{KOH}_{(aq)} \rightarrow \text{KNO}_{3(aq)} + \text{H}_2\text{O}_{(l)}$

4. Refer to the Activity Series of Metals, and indicate reaction (Rxn) or no reaction (NR) when a small piece of **tin metal (Sn)** is dropped into each of the following aqueous solutions.

- (a) $\text{Ca}(\text{NO}_3)_{2(aq)}$ _____ (b) $\text{Cr}(\text{NO}_3)_{3(aq)}$ _____
 (c) $\text{Ni}(\text{NO}_3)_{2(aq)}$ _____ (d) $\text{Pb}(\text{NO}_3)_{2(aq)}$ _____
 (e) $\text{HNO}_{3(aq)}$ _____ (f) $\text{AgNO}_{3(aq)}$ _____

5. Answer the following question using the activity series. If a reaction occurs, **write a balanced molecular, total ionic and net ionic equations**.

- (a) Would silver react with dilute sulfuric acid? Why or why not?
 (b) Would magnesium react with dilute sulfuric acid? Why or why not?

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6.3: Assessment

Choose the Correct Answer: [1 point each]

- Based on the solubility rules, which one of these compounds is *insoluble* in water?
A) Na_2SO_4 B) BaSO_4 C) CuSO_4 D) MgSO_4 E) Rb_2SO_4
- Based on the solubility rules, which one of these compounds is *soluble* in water?
A) AgBr B) AgCl C) Ag_2CO_3 D) AgNO_3 E) Ag_2S
- Based on the solubility rules, which of these processes will occur if solutions of $\text{CuSO}_4(aq)$ and $\text{BaCl}_2(aq)$ are mixed?
A) CuCl_2 will precipitate; Ba^{2+} and SO_4^{2-} are spectator ions.
B) CuSO_4 will precipitate; Ba^{2+} and Cl^- are spectator ions.
C) BaSO_4 will precipitate; Cu^{2+} and Cl^- are spectator ions.
D) BaCl_2 will precipitate; Cu^{2+} and SO_4^{2-} are spectator ions.
E) No precipitate will form.
- Based on the solubility rules, which of these processes will occur when solutions of $\text{ZnSO}_4(aq)$ and $\text{MgCl}_2(aq)$ are mixed?
A) ZnCl_2 will precipitate; Mg^{2+} and SO_4^{2-} are spectator ions.
B) ZnSO_4 will precipitate; Mg^{2+} and Cl^- are spectator ions.
C) MgSO_4 will precipitate; Zn^{2+} and Cl^- are spectator ions.
D) MgCl_2 will precipitate; Zn^{2+} and SO_4^{2-} are spectator ions.
E) No precipitate will form.
- If aqueous solutions of Na_2CO_3 and BaCl_2 are mixed, which insoluble precipitate is formed?
A) Ba_2CO_3 B) BaCO_3 C) NaCl D) NaCl_2 E) BaO
- What is the net ionic equation if sodium sulfate is mixed with barium hydroxide?
A) $2\text{Na}^+(aq) + \text{SO}_4^{2-}(aq) \rightarrow \text{Na}_2\text{SO}_4(s)$
B) $\text{Na}^+(aq) + \text{OH}^-(aq) \rightarrow \text{NaOH}(s)$
C) $2\text{Ba}^+(aq) + \text{SO}_4^{2-}(aq) \rightarrow \text{Ba}_2\text{SO}_4(s)$
D) $\text{Ba}^{2+}(aq) + \text{SO}_4^{2-}(aq) \rightarrow \text{BaSO}_4(s)$
E) $\text{Ba}^{2+}(aq) + 2\text{OH}^-(aq) \rightarrow \text{Ba}(\text{OH})_2(s)$
- Which salt is formed in the neutralization reaction of hydrochloric acid with calcium hydroxide?
A) CaO B) CaCl C) CaH_2 D) CaCl_2 E) NaCl
- Which of the following is a weak acid?
A) H_2SO_4 B) HNO_3 C) HF D) HBr E) HCl
- Which of the following is a strong base?
A) NH_3 B) $\text{Ca}(\text{OH})_2$ C) $\text{Al}(\text{OH})_3$ D) $\text{B}(\text{OH})_3$ E) CH_3OH
- The distinguishing characteristic of all electrolyte solutions is that they
A) contain molecules. B) conduct electricity. C) react with other solutions.
D) always contain acids. E) conduct heat.
- Which of these compounds is a *strong electrolyte*?

A) H_2O B) O_2 C) H_2SO_4 D) $\text{C}_6\text{H}_{12}\text{O}_6$ (glucose) E) CH_3COOH (acetic acid)

12. Which of these compounds is a *weak electrolyte*?

A) HCl B) CH_3COOH (acetic acid) C) $\text{C}_6\text{H}_{12}\text{O}_6$ (glucose) D) O_2 E) NaCl

13. Which of these compounds is a *nonelectrolyte*?

A) NaF B) HNO_3 C) CH_3COOH (acetic acid) D) NaOH E) $\text{C}_6\text{H}_{12}\text{O}_6$ (glucose)

14. Which of these chemical equations describes a *precipitation reaction*?

A) $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{l})$

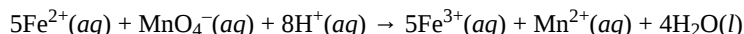
B) $\text{CaBr}_2(\text{aq}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{CaSO}_4(\text{s}) + 2\text{HBr}(\text{g})$

C) $2\text{KNO}_3(\text{s}) \rightarrow 2\text{KNO}_2(\text{s}) + \text{O}_2(\text{g})$

D) $2\text{KBr}(\text{aq}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{KCl}(\text{aq}) + \text{Br}_2(\text{l})$

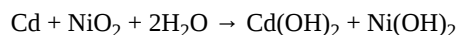
E) $2\text{Al}(\text{s}) + 3\text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{Al}_2(\text{SO}_4)_3(\text{aq}) + 3\text{H}_2(\text{g})$

15. What is the oxidizing agent in the following reaction?



A. H^{+} B. Fe^{2+} C. Fe^{3+} D. MnO_4^{-} E. Mn^{2+}

16. Identify the *reducing agent* in the chemical reaction



A. $\text{Ni}(\text{OH})_2$ B. NiO_2 C. Cd D. H_2O E. $\text{Cd}(\text{OH})_2$

17. Which salt is produced by the neutralization of hydrobromic acid with magnesium hydroxide?

A. Mg_2Br B. MgBr C. Mg_2Br_3 D. MgBr_2 E. Mg_3Br_2

18. Select the compound in which sulfur has its highest possible oxidation number.

A. H_2S B. H_2SO_3 C. Na_2SO_4 D. SO_2 E. SCl_2

19. Which of these chemical equations does not describe a *redox reaction*?

A. $2\text{KNO}_3(\text{s}) \rightarrow 2\text{KNO}_2(\text{s}) + \text{O}_2(\text{g})$

B. $2\text{Na}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{NaOH}(\text{aq}) + \text{H}_2(\text{g})$

C. $3\text{NO}_2(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow 2\text{HNO}_3(\text{aq}) + \text{NO}(\text{g})$

D. $2\text{KBr}(\text{aq}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{KCl}(\text{aq}) + \text{Br}_2(\text{l})$

E. $2\text{LiOH}(\text{aq}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{Li}_2\text{SO}_4(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$

20. Which is a correct description of the following reaction? $\text{H}_2\text{SO}_4 + \text{Cd}(\text{OH})_2 \rightarrow 2\text{H}_2\text{O} + \text{CdSO}_4$

A. In this reaction, oxygen is oxidized.

B. In this reaction, hydrogen is oxidized.

C. This reaction is an acid-base reaction.

D. In this reaction, cadmium is oxidized.

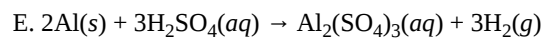
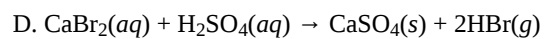
E. In this reaction, sulfur is oxidized.

21. Which equation describes a *redox reaction*?

A. $2\text{KBr}(\text{aq}) + \text{Pb}(\text{NO}_3)_2(\text{aq}) \rightarrow 2\text{KNO}_3(\text{aq}) + \text{PbBr}_2(\text{s})$

B. $\text{H}^{+}(\text{aq}) + \text{OH}^{-}(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$

C. $\text{CO}_3^{2-}(\text{aq}) + \text{HSO}_4^{-}(\text{aq}) \rightarrow \text{HCO}_3^{-}(\text{aq}) + \text{SO}_4^{2-}(\text{aq})$



22. What is the oxidation number of N in N_2H_4 ?

A. -2 B. -4 C. +2 D. 0 E. +4

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